# VECTOR ALZEBRA

→A vector is a quantity that has both magnitude and direction in space.

Ex: Force, Velocity, displacement & acceleration

→A scalar is a quantity that has only magnitude.

Ex: mass, time, temperature and work.

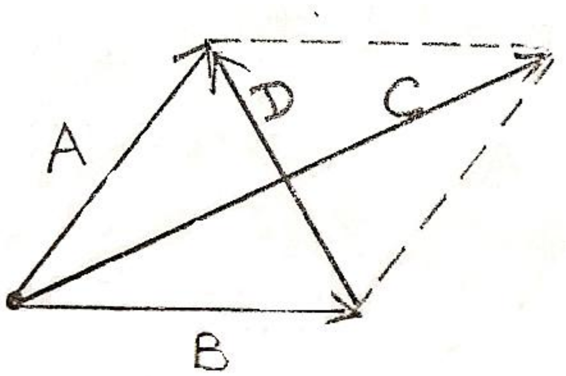
→A unit vector is a vector whose magnitude is unity (i.e:1)
→The vector $\vec{A}$ can be written as

$$\vec{A} = |A|.i_A$$

$$iA = \frac{\vec{A}}{|A|}$$

$i_A$ is the unit vector in the direction of A.
→Two vectors A and B when added and subtracted will give another vector.

Graphically, vector addition and subtraction are obtained by the parallelogram rule.




Vector Addition C = A+B
Vector subtraction D = A-B

**VECTOR MULTIPLICATION:**

→When two vectors A and B are multiplied the result is either a scalar (or) a vector depending on how they are multiplied.
→There are two types of vector multiplication.

- Scalar (or dot) product (A.B)
- Vector (or cross) product (AXB)

$\rightarrow$Multiplication of three vectors can be result either in

i) Scalar triple product A.(BXC)

ii) Vector triple product AX(BXC)

**DOT PRODUCT:**

$\rightarrow$The dot product of two vectors A and B, written as; A.B, is defined geometrically as the product of the magnitude of A and B and cosine of the angle between

Thus: $A.B = |A|\,|B| \cos\theta_{AB}$

$\rightarrow$If A.B=0 then two vectors A and B are said to be orthogonal (or) perpendicular.

If $A = A_x i_x + A_y i_y + A_z i_z$

$B = B_x i_x + B_y i_y + B_z i_z$ then

$A.B = A_x B_x + A_y B_y + A_z B_z$

$\rightarrow$ $A.B = B.A$

$A.A = |A|^r$

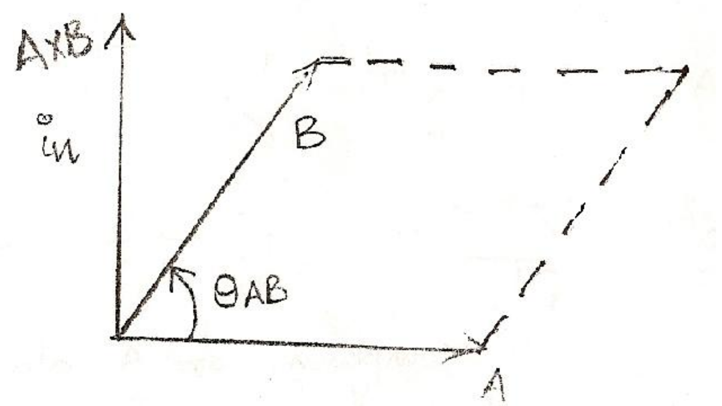
$\rightarrow i_x.i_x = i_y.i_y = i_z.i_z = 1$

$i_x.i_y = i_y.i_z = i_z.i_x = 0$

**CROSS PRODUCT:**

$\rightarrow$The cross product of two vectors A and B written as AXB, is a vector quantity whose magnitude is the area of the parallelopiped formed by A and B and is in the direction of a right hand screw as A is turned into B.

$$AXB = |A|\,.\,|B|.\sin\theta_{AB}$$




$\rightarrow$If $A = A_x i_x + A_y i_y + A_z i_z$ and

$B = B_x i_x + B_y i_y + B_z i_z$ then

$$AXB = \begin{vmatrix} i_x & i_y & i_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$AXB = (A_yB_z - B_yA_z)i_x + (A_zB_x – A_xB_z)i_y + (A_xB_y – B_xA_y)i_z$

$\rightarrow$ $AXB \neq BXA$

$AXB = - BXA$

$AXA = 0$

$\rightarrow$ $i_xXi_x = i_yXi_y = i_zXi_z = 0$

$i_xXi_y = i_z; i_yXi_z = i_x; i_zXi_x = i_y$

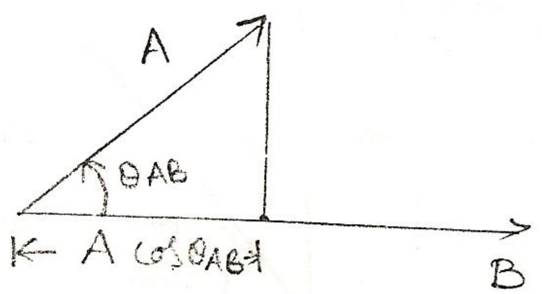
$\rightarrow$A, B and C are the three rectors then the scalar tripple product is

A. (BXC) = B. (CXA) – C. (AXB)

$\rightarrow$The vector tripple product is

AX (BXC) = B. (A.C) – C. (A.B)

**COMPONENT OF A VECTOR**




$\rightarrow$The component of A along B is = A $\cos\theta_{AB}$

$$= \frac{AB\cos\theta_{AB}}{B}$$

$$= \frac{A.B}{B} \qquad \text{(Scalar component)}$$

$$= \frac{(A.B).\vec{B}}{|B|^{r}} \qquad \text{(Vector component)}$$

## Problems:-

1. Given that $A = i_x + \alpha i_y + i_z$ and $B = \alpha i_x + i_y + i_z$ if A and B are normal to each other what is $\alpha$ = ………………….

   $A.B = 0 \qquad [\because A \perp B]$

   $(i_x + i_y + i_z).(\alpha i_x + i_y + i_z) = 0$

   $\alpha + \alpha + 1 = 0$

   $\alpha\alpha + 1 = 0$

   $\alpha = -1/2$

2. Given $A = -6i_x+3i_y+2i_z$ the projection of A along $i_y$ is

a) -12 b) -4 c) 3 d) 12

Soln: $A.i_y = 3$

3. The component of $A = 6i_x+2i_y-3i_z$ along $B = 3i_x-4i_y$ is

Soln: The component of A along B is = A $\cos\theta_{AB}$

$$= \frac{A.B}{|B|} = \frac{(6i_x + 2i_y + -3i_z)(3i_x - 4i_y)}{\sqrt{(3)^2 + (-4)^2}}$$

$$= \frac{18-8}{\sqrt{25}} = \frac{10}{5} = 2$$

4. Given vectors

$A = \alpha ix+iy+4iz$

$B = 3ix+\beta iy-6iz$

$C = 5ix-2iy+\gamma iz$

Determine $\alpha$,$\beta$ and $\gamma$ such that the vectors are mutually orthogonal.

Soln: A.B = 0, B.C = 0, C.A = 0 $[\because A \perp B \perp C]$

$A.B = 0 \Rightarrow 3\alpha+\beta-24 = 0 \Rightarrow 3\alpha+\beta = 24 \rightarrow (1)$

$B.C = 0 \Rightarrow 15+2\beta-6\gamma = 0 \Rightarrow 2\beta+6\gamma = 15 \rightarrow (2)$

$C.A = 0 \Rightarrow 5\alpha-2+4\gamma = 0 \Rightarrow 5\alpha+4\gamma = 2 \rightarrow (3)$

From (1) $\beta = 24-3\alpha$

Substituting in eqn.(2)

$2(24-3\alpha) +6\gamma = 15$

$6(8-\alpha) +6\gamma = 15$

$2(8-\alpha) +2\gamma = 5$

$16-2\alpha+2\gamma = 5$

$-2\alpha+2\gamma = 5-16$

$-2\alpha+2\gamma = -11$

$\alpha-\gamma = 11/2 \rightarrow (4)$

$4\alpha - 4\gamma = 22$

$5\alpha - 4\gamma = 2$

$9\alpha = 24$

$\alpha = 24/9$

$$\gamma = \frac{24}{9} - \frac{11}{2} = \frac{48-99}{18} = \frac{51}{18} = \frac{17}{6}$$

$$\beta = 24 - 3X\frac{24}{9} = 24 - 8 = 16$$

5. Simplify the following:

a) AX (AXB) b) AX [AX(AXB)]

Soln: a) AX (AXB)

A (A.B)-B (A.A)

b) AX [AX (AXB)]

AX [A (A.B)-B (A.A)]

(AXA)(A.B)- (AXB) (A.A)

# -:CO-ORDINATE – SYSTEMS:-

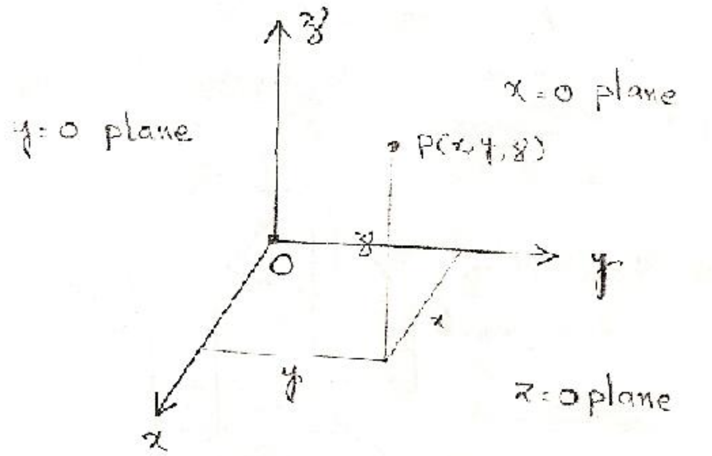
→In order to describe spatial variation of the Electro magnetic waves as a function of space and time, we require using an appropriate co-ordinate system.

→The three simple co-ordinate system's are:

i) Cartesian (Rectangular) co-ordinate system.
ii) Cylindrical co-ordinate system.
iii) Spherical co-ordinate system.

## →Cartesian co-ordinate system:-

→In the Cartesian co-ordinate system, we set up three co-ordinate axes mutually at right angles to each other, and call them, the x, y and z axes.




→A point in catesian co-ordinate system is located by giving its x, y and z co-ordinates these are the distances from the origin to the intersection of a perpendicular dropped from the point to the x, y, and z axes.

→The ranges of the coordinate variables x, y and z are

$-\alpha < x < \alpha$

$-\alpha < y < \alpha$

$-\alpha < z < \alpha$

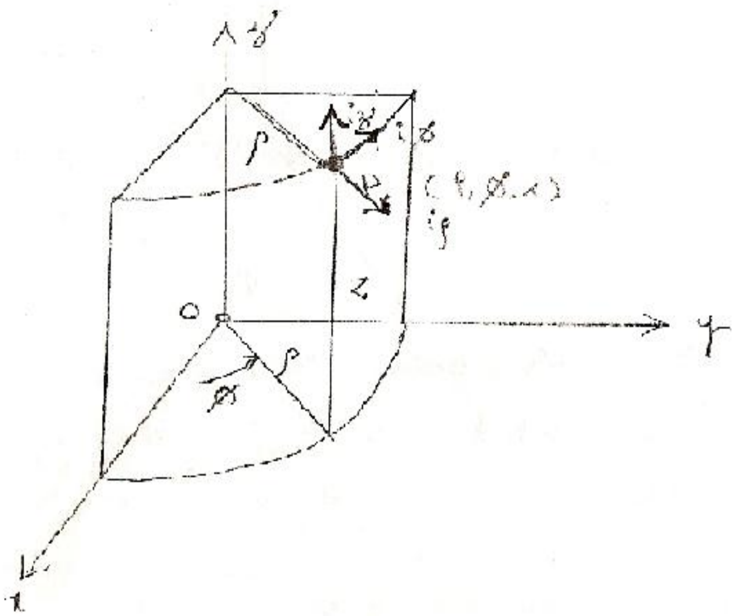
$\rightarrow$A vector $\vec{A}$ in Cartesian co-ordinates can be written as

$$\vec{A} = A_x i_x + A_y i_y + A_z i_z$$

Where $i_x$,$i_y$ and $i_z$ are the unit vectors along x,y and z directions.

# -: CYLINDRICAL CO-ORDINATE SYSTEM:-

$\rightarrow$A point P in cylindrical co-ordinates is represented as (ρ, Φ, z).
Where ρ is the radius of the cylinder
Φ is the azimuth angle, is measured from the x-axis in the xy-plane.



$\rightarrow$The ranges of the co-ordinate variables ρ, Φ and z are

$$0 \le P \le \alpha$$

$$0 \le \phi < 2\pi$$

$$0 < Z < \alpha$$

$\rightarrow$A vector $\vec{A}$ in cylindrical co-ordinates can be written as

$\rightarrow$ ip.ip = $i\phi . i\phi$ = iz.iz = 1

Where ip, $i\phi$ and iz are the unit vectors along $\rho, \phi$ and z directions.

$\rightarrow$ ip.ip = $i\phi . i\phi$ = iz.iz = 1

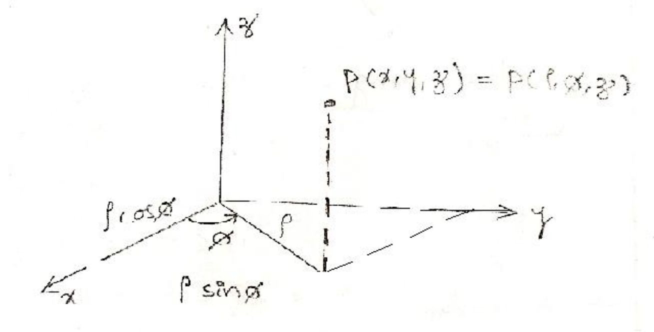
ip.$i\phi$ = $i\phi$.iz = iz.ip = 0

ip X $i\phi$ = iz

$i\phi$ X iz = ip

iz X ip = $i\phi$

## -: Conversion relations b/w Cartesian (x, y, z) and cylindrical (p, $\phi$, z):-



$x = \rho \cos\phi$ $\qquad$ $P = \sqrt{x^2 + y^2}$

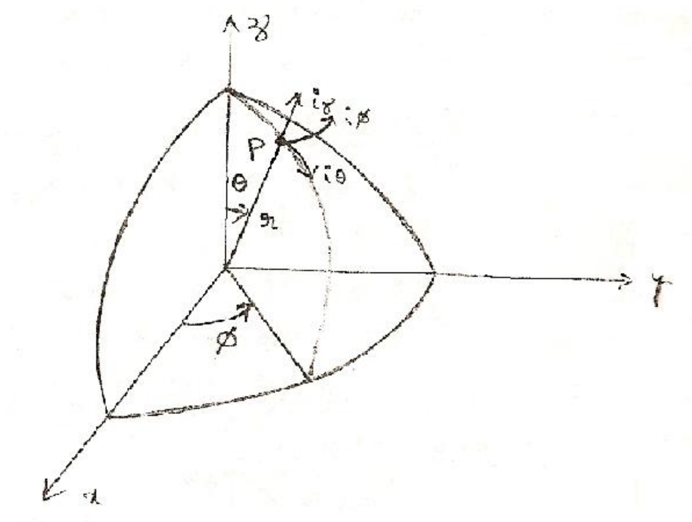
$y = \rho \sin\phi$ $\qquad$ $\phi = \tan^{-1}(y/x)$

$z = z$ $\qquad$ $z = z$

## -: Spherical co-ordinates:-

$\rightarrow$A point P in spherical co-ordinates is represented as $(r, \theta, \phi)$.

Where r is the radius of the sphere. $\theta$ Is the elevation angle and is the angle between z-axis and position vector of $\phi$ is the azimuth angle, is measured from the x-axis.



$\rightarrow$The ranges of the co-ordinate variables r, $\theta$ and $\phi$ are

$$0 \le r < \alpha$$
$$0 \le \theta \le \pi$$
$$0 \le \phi < 2\pi$$

$\rightarrow$A vector Ain spherical co-ordinate may be written as

$$\vec{A} = A_r i_r + A_\theta i_\theta + A_\phi i_\phi$$

Where $i_r$, $i_\theta$ and $i_\phi$ are unit vectors along the r, θ and $\phi$-directions.

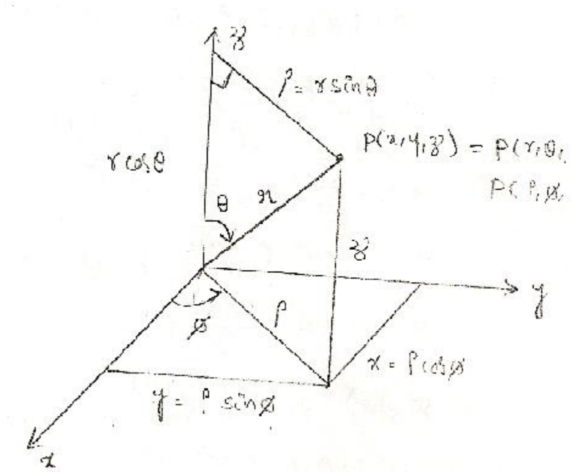
$\rightarrow$ $i_r.i_r = i_\theta.i_\theta = i_\phi.i_\phi = 1$

$i_r.i_\theta = i_\theta.i_\theta = i_\phi.i_r = 0$

$i_r X i_\theta = i_\phi$ $\qquad$ $i_\theta X i_\phi = i_r$

$i_\phi X i_r = i_\theta$

## <u>-: Conversion Relations b/w Cartesian (x, y, z) and Spherical(r,θ,$\phi$):-</u>




$x = r\sin\theta.\cos\phi$ $\qquad$ $r = \sqrt{x^2 + y^2 + z^2}$

$y = r\sin\theta.\sin\phi$ $\qquad$ $\theta = \tan^{-1}\dfrac{\sqrt{x^2+y^2}}{z}$

$z = r\cos\theta$ $\qquad$ $\phi = \tan^{-1}(y/x)$

Ex:1) Convert a point P(1,3,5) from Cartesian to cylindrical and spherical co-ordinates.

Soln: P (1, 3, 5)

$x = 1, y = 3, z = 5$

$$\rho = \sqrt{x^2 + y^2} = \sqrt{1+9} = \sqrt{10} = 3.162$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(3) = 71.56^o$$

$z = 5$

$P(p, \phi, z) = (3.162, 71.56^o, 5)$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + 9 + 25}$$
$$r = \sqrt{35} \quad = \quad 5.916$$
$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$$
$$\theta = \tan^{-1}\left(\frac{3.162}{5}\right) = 32.31^o$$
$$\phi = \tan{-1}\left(\frac{y}{x}\right) = 71.56^o$$
$$P(r, \theta, \phi) = P(5.916, 32.31^o, 71.56^o)$$

# Vector transformation from Rectangular to Cylindrical

$\rightarrow$A vector $\vec{A}$ in rectangular co-ordinate system can be represented as

$$\vec{A} = A_x i_x + A_y i_y + A_z i_z$$

$\rightarrow$And it can be represented in cylindrical coordinate system as

$$\vec{A} = A_p i_p + A_\phi i_\phi + A_z i_z$$

To find $A_p$

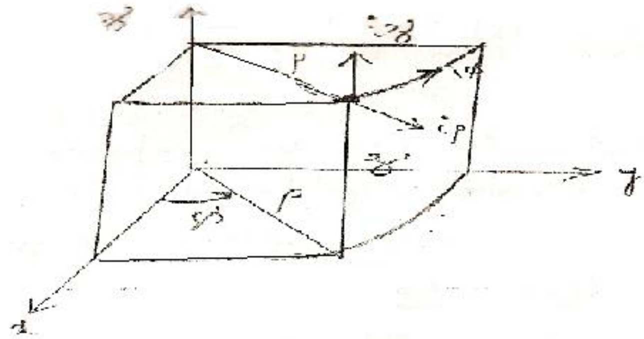
$$A_p = \vec{A}.i_p$$
$$= (A_x i_x + A_y i_y + A_z i_z).i_p$$
$$= A_x(i_x.i_p) + A_y(i_y.i_p) + A_z(i_z.i_p)$$
$$= A_x.\cos\phi + A_y.\sin\phi$$

To find $A_\phi$:

$$A_\phi = \vec{A}.i_\phi$$
$$= (A_x i_x + A_y i_y + A_z i_z).i_\phi$$
$$= A_x(i_x.i_\phi) + A_y(i_y.i_\phi) + A_z(i_z.i_\phi)$$
$$= A_x(-\sin\phi) + A_y(\cos\phi)$$

To find $A_z$:

$$A_z = \vec{A}.i_z$$
$$= (A_x i_x + A_y i_y + A_z i_z).i_z$$
$$= A_x X 0 + A_y X 0 + A_z$$
$$= A_z$$



| . | $i_\rho$ | $i_\phi$ | $i_z$ |
|---|---|---|---|
| $i_x$ | $\cos\phi$ | $-\sin\phi$ | 0 |
| $i_y$ | $\sin\phi$ | $\cos\phi$ | 0 |
| $i_z$ | 0 | 0 | 1 |

$$\therefore \vec{A} = \left[A_x\cos\phi + A_y\sin\phi\right]i_p + \left[-A_x\sin\phi + A_y\cos\phi\right] + A_z i_z$$

* * *

$$\begin{bmatrix} A_p \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

## -: Vector transformation from Rectangular to Spherical:-

$\rightarrow$A vector $\vec{A}$ in rectangular co-ordinate system can be represented as:

$$\vec{A} = A_x i_x + A_y i_y + A_z i_z$$

$\rightarrow$And it can be represented in spherical co-ordinate system as:

$$\vec{A} = A_r i_r + A_\theta i_\theta + A_\phi i_\phi$$

<u>To find $A_r$:-</u>

$$A_r = \vec{A}.i_r$$

$$= (A_r i_r + A_\theta i_\theta + A_\phi i_\phi).i_r$$

$$= A_x(i_x.i_r) + A_y(i_y.i_r) + A_z(i_z.i_r)$$

$$= A_x(\sin\theta\cos\phi) + A_y(\sin\theta\sin\phi) + A_z(\cos\theta)$$

$$A_r = Ax\sin\theta\cos\phi + Ay\sin\theta\sin\phi + Az\cos\theta$$

<u>To find $A_\theta$:-</u>

$$A_\theta = \vec{A}.i_\theta$$

$$= (A_x i_x + A_y i_y + A_z i_z).i_\theta$$

$$= A_x(i_x.i_\theta) + A_y(i_y.i_\theta) + A_z(i_z.i_\theta)$$

$$= A_x[\cos\theta\cos\phi] + A_y[\cos\theta\sin\phi] + A_z[-\sin\theta]$$

$$A_\theta = A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta$$

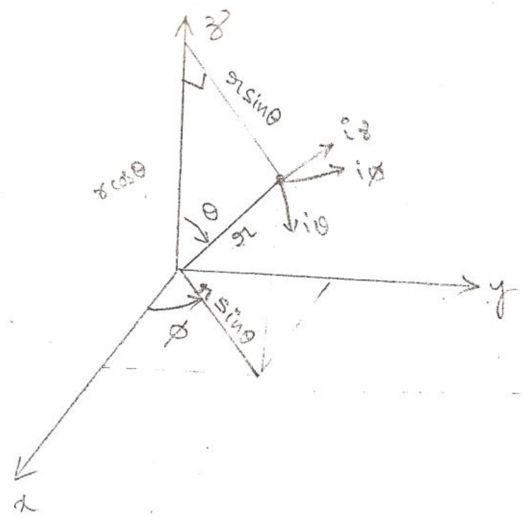
<u>To find</u> $\underline{\overline{A_\phi}:}$

$$A_\phi = \vec{A}.i_\phi$$

$$A_\phi = (A_x i_x + A_y i_y + A_z i_z).i_\phi$$

$$A_\phi = A_x(i_x.i_\phi) + A_y(i_y.i_\phi) + A_z(i_z.i_\phi)$$

$$A_\phi = -A_x\sin\phi + A_y\cos\phi$$




| . | $i_x$ | $i_y$ | $i_z$ |
|---|---|---|---|
| $i_r$ | $\sin\theta\cos\phi$ | $\sin\theta\sin\phi$ | $\cos\theta$ |
| $i\theta$ | $\cos\theta\cos\phi$ | $\cos\theta\sin\phi$ | $-\sin\theta$ |
| $i_\phi$ | $-\sin\phi$ | $\cos\phi$ | 0 |

$$\therefore \vec{A} = (A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta)i_r + (A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta)i_\theta$$

$$+(-A_x\sin_\phi + A_y\cos_\phi)i_\phi$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

# -:Constant co-ordinate surfaces:-

→Surfaces in Cartesian, cylindrical (or) Spherical co-ordinate systems are easily generated by keeping one of the co-ordinate variables constant and allowing the other two to vary.

→In Cartesian co-ordinate system we would have three infinite planes.

$$x = \text{constant}$$
$$y = \text{constant}$$
$$z = \text{constant}$$

→The orthogonal surfaces in cylindrical co-ordinates are:

$\rho$ = constant, is a circular cylinder

$\phi$ = constant, is a semi infinite plane

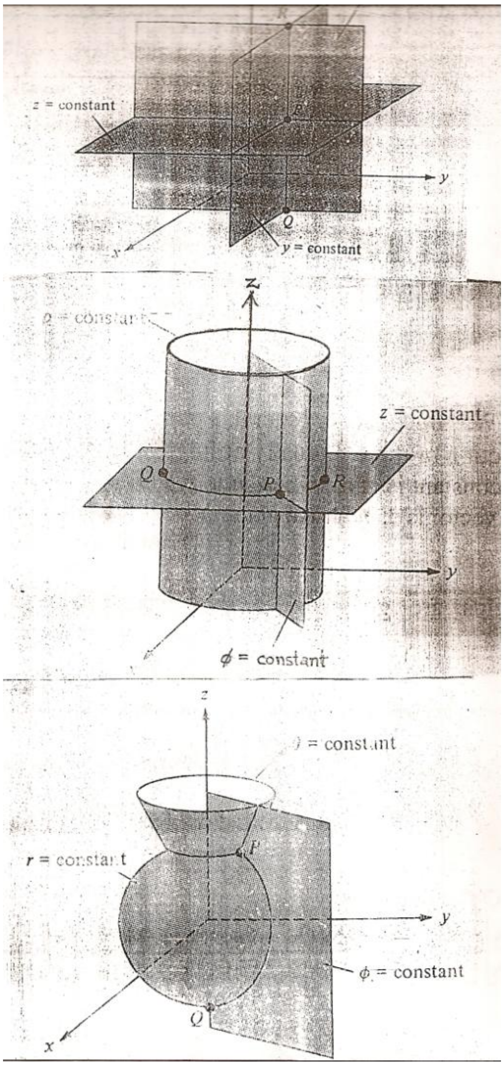
z = constant, is a infinite plane

→The orthogonal surfaces in spherical co-ordinates are:

r = constant, is a sphere with its center at the origin

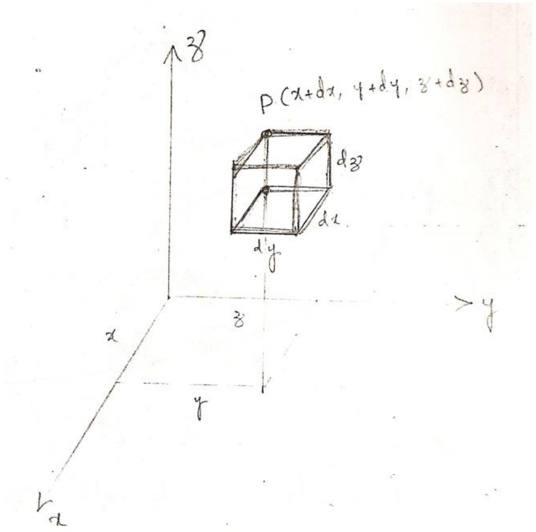
$\theta$ = constant, is a circular cone with the z-axis as its axis

$\phi$ = constant, is a semi infinite plane



# -:Differential Length, Area and Volulme:-

## Cartesian co-ordinates:-




→Differential displacement is given by

dl = dxix+dyiy+dziz

→Differential normal area is given by

dS = $d_y d_z$ $i_x$

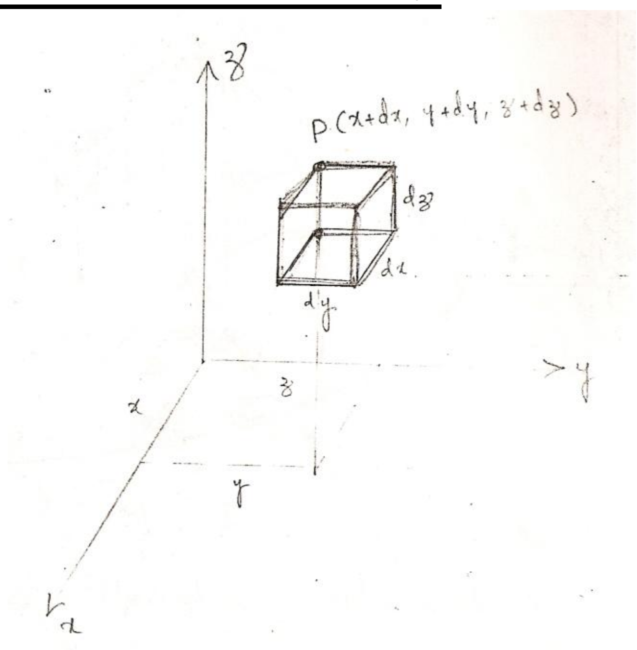
= $d_z d_x$ $i_y$

= $d_x d_y$ $i_z$

→Differential volume is given by

dV = dx.dy.dz

## Cylindrical co-ordinates:-



$\rightarrow$Differential displacement is given by

$$dl = d\rho i_\rho + \rho d\phi i_\phi + dz i_z$$

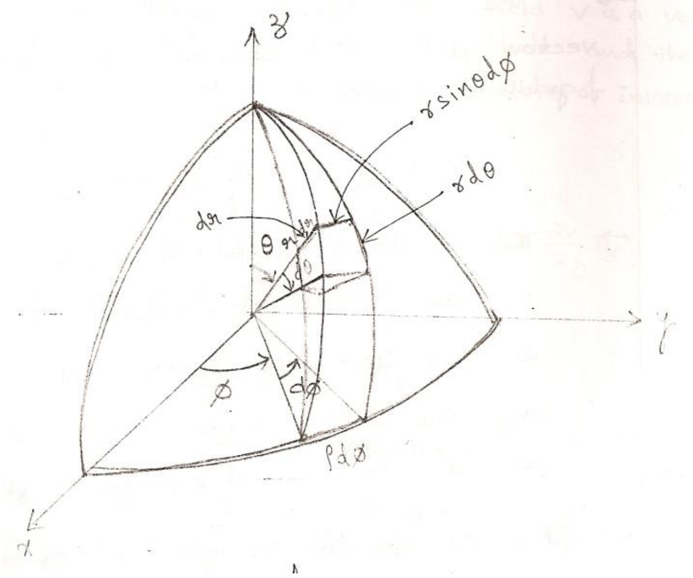
$\rightarrow$Differential normal area is given by

$$ds = \rho d\phi dz.i_p$$
$$= d\rho dz.i_\phi$$
$$= \rho d\phi d\rho.i_z$$

$\rightarrow$Differential volume is given by

$$dV = \rho d\rho d\phi dz$$

# Spherical co-ordinates:-




$\rightarrow$The differential displacement is

$$dl = dri_r + rd\theta i\theta + r\sin\theta d\phi\, i_\phi$$

$\rightarrow$The differential normal area is

$$dS = r^2\sin\theta\, d\theta\, d\phi\ \ i_r$$
$$= r\sin\theta\, dr\, d\phi\ \ i_\theta$$
$$= r\, dr\, d\theta\ i_\phi$$

$\rightarrow$The differential volume is

$$dV = r^2 \sin\theta\, dr.d\theta.d\phi$$

## -: DEL OPERATOR:-

$\rightarrow$The del operator, written as $\nabla$, is the vector differential operator known as the gradient operator.

$$\nabla = \frac{\partial}{\partial x} i_x + \frac{\partial}{\partial y} i_y + \frac{\partial}{\partial z} i_z \text{ (In Cartesian co-ordinates)}$$

$\rightarrow$The operator is useful in defining

1. The gradient of a scalar, $\nabla V$
2. The divergence of a vector A, $\nabla . A$
3. The curl of a vector A, $\nabla X A$
4. The Laplacian of a scalar V, $\nabla^2 V$

$\rightarrow$The $\nabla$, operator in cylindrical co-ordinates as:

$$\nabla = \frac{\partial}{\partial \rho} i_p + \frac{1}{\rho} . \frac{\partial}{\partial \phi} i_\phi + \frac{\partial}{\partial z} i_z$$

$\rightarrow$The $\nabla$ operator in spherical co-ordinates as:

$$\nabla = \frac{\partial}{\partial r} i_r + \frac{1}{r} \frac{\partial}{\partial \theta} i_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} i_\phi$$

# : GRADIENT OF A SCALAR:

$\rightarrow$The gradient of a scalar field V is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V.

$\rightarrow$For Cartesian co-ordinates:

$$\nabla V = \text{grad } V = \frac{\partial V}{\partial x} i_x + \frac{\partial V}{\partial y} i_y + \frac{\partial V}{\partial z} i_z$$

$\rightarrow$For cylindrical co-ordinates:

$$\nabla V = \frac{\partial V}{\partial \rho} i_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} i_\phi + \frac{\partial V}{\partial z} i_z$$

$\rightarrow$For spherical co-ordinates:

$$\nabla V = \frac{\partial V}{\partial r} i_r + \frac{1}{r} \frac{\partial V}{\partial \theta} i_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} i_\phi$$

$\rightarrow$The fundamental properties of the gradient of a scalar field V is:

i) The magnitude of $\nabla V$ equals the maximum rate of change of V per unit distance.

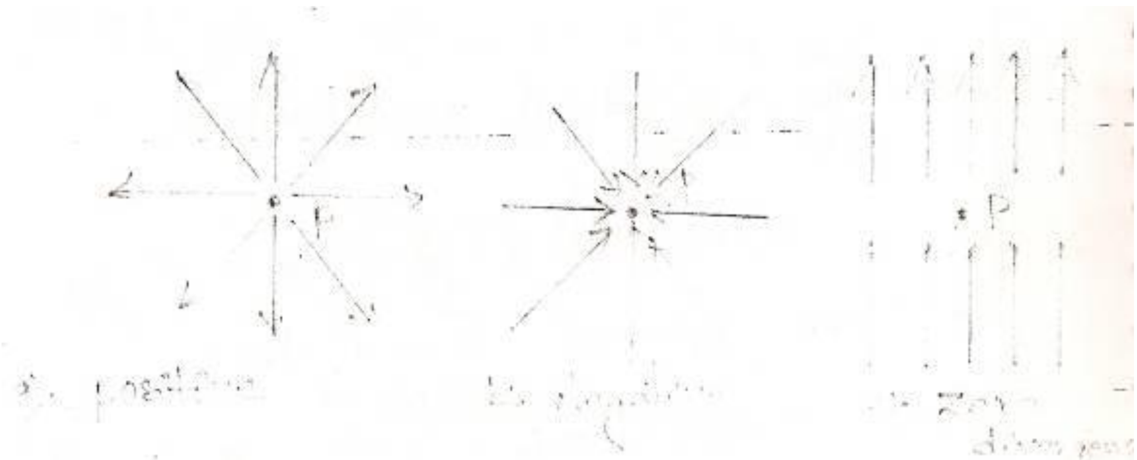
ii) $\nabla V$ points in the direction of the maximum rate of change in V.

iii) $\nabla V$ at any point is perpendicular to the constant V surface that passes through that point.

# -: Divergence of a vector:-

$\rightarrow$The divergence of a vector field A at a given point is the outward flux per unit volume as the volume shrinks about P.

Hence div A = $\nabla.A = \underset{\Delta V \to 0}{Lt} \dfrac{\oint_S A.ds}{\Delta V}$



$\rightarrow$In fig (a) shows that the divergence of a vector field at point P is positive because the vector diverges (or) spreads out at P.

$\rightarrow$In fig (b) A vector field has negative divergence a convergence at P.

$\rightarrow$In fig(c) A vector field has zero divergence.

$\rightarrow$Divergence of a vector field has the following properties.

1. It producer a scalar field.
2. The divergence of a scalar makes no sense.
3. $\nabla.(A+B) = \nabla.A + \nabla.B$
4. $\nabla.(AV) = V\nabla.A + A.\nabla V$

$\rightarrow$The divergence of a vector in Cartesian co-ordinates is given by

$$\nabla.A = \frac{\partial}{\partial x} A_x + \frac{\partial}{\partial y} A_y + \frac{\partial}{\partial z} A_z$$

$\rightarrow$The divergence of a vector in cylindrical co-ordinates is:

$$\nabla.A = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}.\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$\rightarrow$The divergence of a vector in spherical co-ordinates is:

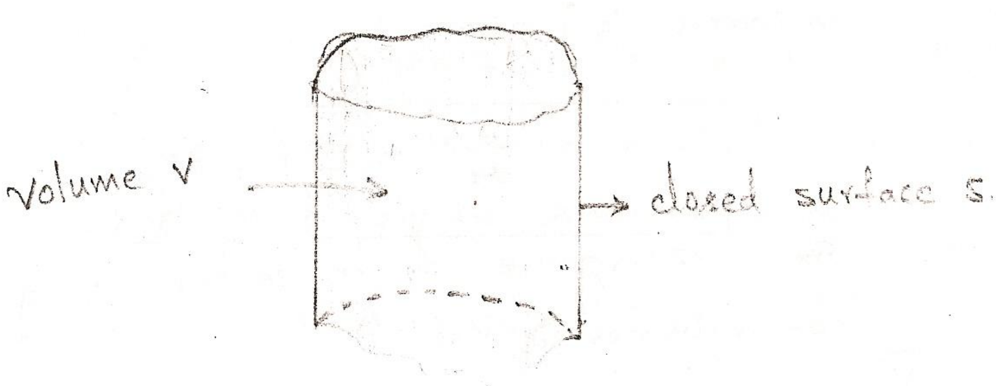
$$\nabla.A = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2 A_r\right] + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}[A_\theta \sin\theta] + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial \phi}$$

# -: Divergence Theorem:-
# -: Gauss-Ostrogradsky:-

$\rightarrow$The divergence theorem states that " the total out ward flux of a vector field A through the closed surface S is the same as the volume integral of the divergence of A".

$$\oint_S A.ds = \int_V \nabla.A\, dv$$

Proof: - Consider a volume V closed by a surface S.




$\rightarrow$To prove the divergence theorem, subdivide volume V into a large number of small cells. If the $K^{th}$ cell has volume $\Delta V_K$ and is bounded by surface $S_K$.

$$\oint_S A.ds = \sum_K \oint_{S_K} A.ds = \sum_K \frac{\oint_{S_K} A.ds}{\Delta V_K}.\Delta V$$

$\rightarrow$Since the outward flux to one cell is inward to some neighboring cells, there is a cancellation on every interior surface, so the sum of the surface integrals over $S_K$'s is the same as the surface integral over the surface S.

$$\therefore \oint_S A.ds = \int_V \nabla.A\, dv$$

$\rightarrow$This theorem applies to any volume V bounded by the closed surface S.

***

# -: Curl of a vector:-

$\rightarrow$The curl of A is an axial (or) rotational vector whose magnitude is the maximum circulation of A per unit area as the area tends to zero and whose direction is the normal direction of the area.

i.e. Curl A = $\nabla XA = \left( \underset{\Delta S \to 0}{Lt} \frac{\oint_L A.dl}{\Delta S} \right) in_{max}$

$\rightarrow \oint_L A.dl \rightarrow$ The circulation of a vector field A around a closed path L.

$\rightarrow$In Cartesian co-ordinates the Curl of A is easily found using.

$$\nabla XA = \begin{vmatrix} i_x & i_y & i_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$\rightarrow$In cylindrical co-ordinates the curl of A is easily found using

$$\Delta XA = \frac{1}{\rho} \begin{vmatrix} i_\rho & \rho i_\phi & i_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$\rightarrow$In spherical co-ordinates the curl of A is easily found using.

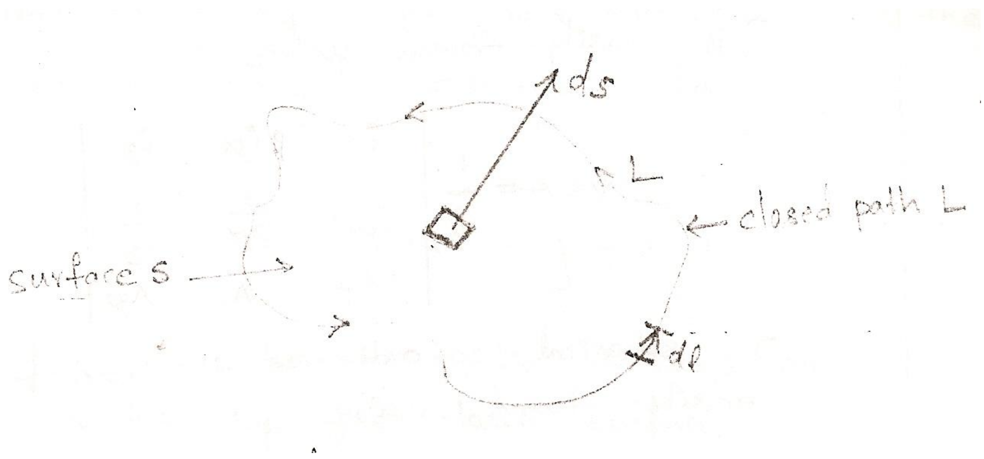
$$\nabla XA = \frac{1}{r^2 \sin\theta} \begin{vmatrix} i_r & ri_\theta & r\sin\theta i_\phi \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

# -: The properties of the curl:-

1. The curl of a vector field is another vector field.
2. The curl of a scalar field makes no sense.
3. $\nabla X(A+B) = \nabla XA + \nabla XB$
4. $\nabla X(AXB) = A(\nabla.B) - B(\nabla.A) + (B.\nabla)A - (A.\nabla)B$
5. $\nabla X(VA) = V(\nabla XA) + \nabla VXA$
6. $\nabla . (\nabla XA)=0$
7. $\nabla X \nabla V = 0$

# -: Stoke's Theorem:-

→Stoke's theorem states that the circulation of a vector field A around a closed path L is equal to the surface integral of the curl of A over the open surface bounded by L.




$$\oint_L A.dl = \int_S (\nabla X A).ds$$

Proof:
→The proof of Stoke's theorem is similar to that of the divergence theorem. The surface S is subdivided into a large number of cells. If the $K_{th}$ cell has surface area $\Delta S_K$ and is bounded by path $L_K$

$$\rightarrow \oint_L A.dl = \sum_K \oint_{L_K} A.dl = \sum_K \frac{\int_{L_K} A.dl}{\Delta S_K}.\Delta S_K$$

There is a cancellation on every interior path, so the sum of the Line integrals around the $L_K$'s is the same as the Line integral around the bounding curve L.

$$\therefore \oint_L A.dl = \int_S (\nabla X A).ds$$

***

# -: Laplacian of a scalar:-

$\rightarrow$The Laplacian of a scalar field 'V', written as $\nabla^2 V$ is the divergence of the gradient of Laplacian V= $\nabla.\nabla V = \nabla^2 V$

$\rightarrow$Laplacian of a scalar field is another scalar field.

$\rightarrow$In Cartecian co-ordinates:-

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$\rightarrow$In cylindrical co-ordinates:-

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$\rightarrow$In spherical co-ordinates:-

$$\nabla^2 V = \frac{1}{r^2}.\frac{\partial}{\partial r}\left(r^2.\frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}.\frac{\partial^2 V}{\partial \phi^2}$$

$\rightarrow$If $\nabla^2 V = 0$, a scalar field 'V' is said to be <u>harmonic</u> in a given region

$\rightarrow \nabla^2$ is a scalar operator.

$\rightarrow$Laplacian of vector field 'A' is given by

$$\nabla^2 A = \nabla(\nabla.A) - \nabla X(\nabla XA)$$

## -: Classification of vector fields:-

$\rightarrow$A vector field 'A' is said to be <u>Solenoidal</u> (or) <u>divergence less</u> if $\nabla.A = 0$.

$\rightarrow$A vector field 'A' is said to be <u>irrotational</u> (or) <u>potential</u> if $\nabla XA = 0$.

# -: Coulomb's Law:-

$\rightarrow$Let two point charges $Q_1$ and $Q_2$ are separated in a vaccum (or) free space by a distance 'R' meters.

$\rightarrow$Coulomb's Law states that the magnitude of force between the two point charges $Q_1$ and $Q_2$ is

(i) Directly proportional to the product $Q_1$, Q2 of charges.

$$F \alpha Q_1 Q$$

(ii) Inversely proportional to the square of the distance 'R' between them.

$$F \alpha \frac{1}{R^2}$$

$\rightarrow$The direction of the force is along the line joining the two charges.

$$F \alpha \frac{Q_1 Q_2}{R^2}$$

$$F = K.\frac{Q_1 Q_2}{R^2} \quad \rightarrow (1)$$

Where K is the proportionality constant.

$\rightarrow$In SI Units, charges $Q_1$ and $Q_2$ are in Coulomb, the distance R is in meters and the force F is in newtons.

So that $K = \frac{1}{4\pi \in_0}$

The constant $\in_0$ is known as the permittivity of free space in Farads/meter, and has the value

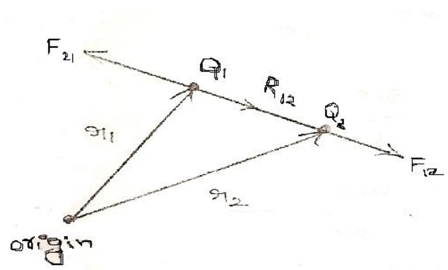
$$\in_0 = 8.854 X 10^{-12} \approx \frac{10^{-9}}{36\pi} F/m$$

(or) $K = \frac{1}{4\pi \in_0} = \frac{1}{4\pi X \frac{10^{-9}}{36\pi}} = 9X10^9 \, m/F$

$\rightarrow$ $\therefore$ eqn(1) becomes

$$F = \frac{Q_1 Q_2}{4\pi \in_0 R^2}.i_R$$

$\rightarrow$If point charges $Q_1$ and $Q_2$ are located at points having position vectors $r_1$ and $r_2$, then the force $F_{12}$ on $Q_2$ due to $Q_1$ is given by




$$F_{12} = \frac{Q_1 Q_2}{4\pi \in_0 R^2}.iR_{12}$$

$$R_{12} = r_2 - r_1$$
$$R = |R_{12}|$$
$$iR_{12} = \frac{R_{12}}{|R_{12}|}$$
$$F_{12} = \frac{Q_1 Q_2}{4\pi \in_0 R^3}.R_{12}$$

$$F_{12} = \frac{Q_1 Q_2 (r_2 - r_1)}{4\pi \in_0 |r_2 - r_1|^3}$$

Note:

i) F12 = -F21

ii) Like charges repel each other, while unlike charges attract.

iii) The distance R between the charged bodies $Q_1$ and $Q_2$ must be large compared with the linear dimensions of the bodies.

iv) $Q_1$ and $Q_2$ must be static.

$\rightarrow$If we have more than two point charges, we can use the principle of superposition to determine the force on a particular charge.

If there are N charges $Q_1$, Q2, Q3…….. $Q_N$ located, respectively, at points with position vectors $r_1, r_2, r_3 \ldots\ldots r_n$, the resultant force F on a charge Q located at point r is the vector sum of the forces exerted on Q by each of the charges $Q_1, Q_2, Q_3 \ldots\ldots Q_N$. H

$$F = \frac{QQ_1(r - r_1)}{4\pi \in_0 |r - r_1|^3} + \frac{QQ_2(r - r_2)}{4\pi \in_0 |r - r_2|^3} + \ldots\ldots + \frac{QQ_N(r - r_N)}{4\pi \in_0 |r - r_N|^3}$$

(or)

$$F = \frac{Q}{4\pi \in_0} \sum_{K=1}^{N} \frac{Q_K (r - r_K)}{r - r_K{}^3}$$

"One coulomb is approximately equivalent to $6X10^{18}$ electrons".

# -: Electric field intensity:-

$\rightarrow$The electric field intensity (or) electric field strength E is the force per unit charge.

$$E = \frac{F}{Q}$$

$\rightarrow$The electric field intensity E is in the direction of the Force.

$\rightarrow$The units for electric field intensity E is newton/coulomb (or) Volts/meter.

$\rightarrow$The electric field intensity at a point 'r' due to a point charage 'Q' located at $r^1$ is given by

$$E = \frac{Q}{4\pi \in_0 R^2} . i_r$$

$$E = \frac{Q(r - r^1)}{4\pi \in_0 |r - r^1|^3}$$

$\rightarrow$For N point charges $Q_1, Q_2 \ldots\ldots Q_N$ located at $r_1, r_2, \ldots\ldots r_n$, the electric field intensity at point r is obtained by

$$E = \frac{Q_1(r - r_1)}{4\pi \in_0 |r - r_1|^3} + \frac{Q_2(r - r_2)}{4\pi \in_0 |r - r_2|^3} + \ldots\ldots + \frac{Q_N(r - r_N)}{4\pi \in_0 |r - r_N|^3}$$

(or)

$$E = \frac{1}{4\pi \in_0} \sum_{K=1}^{N} \frac{Q_K (r - r_K)}{|r - r_K|^3}$$

1. Point charages 1 mc and -2 mc are located at (3,2,-1) and (-1,-1,4) respectively. Calculate the electric force on a 10 mc charge located at (0,3,1) and the electric field intensity at that point.

Soln: $F = \sum_{K=1,2} \frac{QQ_K (r - r_K)}{4\pi \in_0 |r - r_K|^3}$

$$F = \frac{Q}{4\pi \in_0} \frac{(r - r_1)Q_1}{|r - r_1|^3} + \frac{(r - r_2)Q_2}{|r - r_2|^3}$$

$$F = \frac{10X10^{-9}}{4\pi X \frac{10^{-9}}{36\pi}} \left[ \frac{[(0,3,1) - (3,2,-1)]10^{-3}}{|-3,1,2|^{3/2}} + \frac{[(0,3,1) - (-1,-1,4)]10^{-2}}{|1,4,-3|^{3/2}} \right]$$

$$F = 10^{-2} X9 \left[ \frac{(-3,1,2)}{(9+1+4)^{3/2}} + \frac{2(1,4,-3)}{(1+16+9)^{3/2}} \right]$$

$F = -6.507\ i_x - 3.817\ i_y + 7.506\ i_z$ mN

At that point

$$E = \frac{F}{Q}$$

$$E = [-6.507 i_x - 3.817 i_y + 7.506 i_z] X \frac{10^{-}}{10X10^{-C}}$$

$$E = -650.7 i_x - 381.7 i_y + 750.6 i_z \ \ KV/m$$

2. Point charges 5 nc and -2 nc are located at (2,0,4) and (-3,0,5) respectively.

a) Determine the force on a 1 n-c. Point charge located at (1,-3,7).

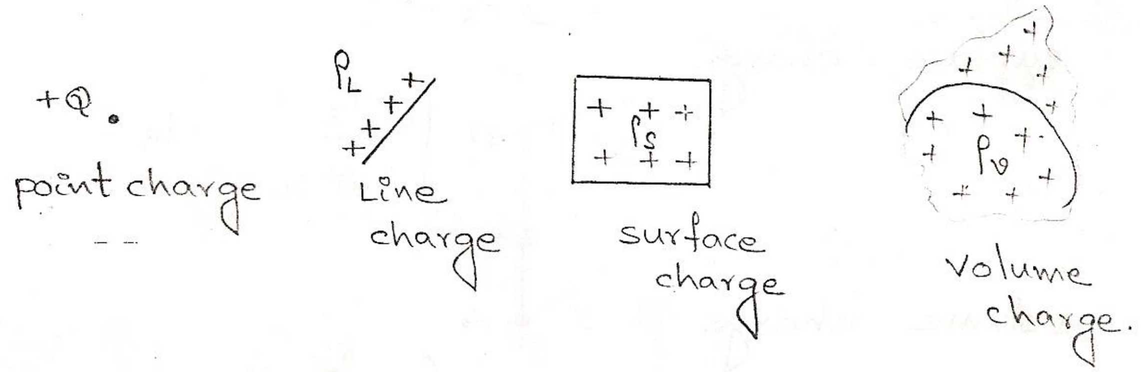
b) Find the electric field E at (1,-3,7).

Soln: Ans: a) $-1.004\ i_x - 1.284\ i_y + 1.4\ i_z$

b) $-1.004\ i_x - 1.284\ i_y + 1.4\ i_z$

# -: Continuous charge distributions:-

→It is also possible to have continuous charge distribution along a line, on a surface or in a volume.




→The Line charge density, surface charge density, and volume charge density are denoted by $\rho_L (c/m)$, $\rho_S (c/m^2)$, $\rho_\vartheta (c/m^3)$, respectively.

→The charge element dQ due to the Line charge distribution is

$$dQ = \rho_L . dl$$

The total charge Q is given by

$$Q = \int_L \rho_L . dl \qquad \text{(Line charge)}$$

→ $dQ = \rho_S . ds$ → $Q = \int_S \rho_S . ds$ (Surface charge)

→ $dQ = \rho_v . dv$ → $Q = \int_v \rho_v . dv$ (Volume charge)

→ The electric field intensity due to Line charge

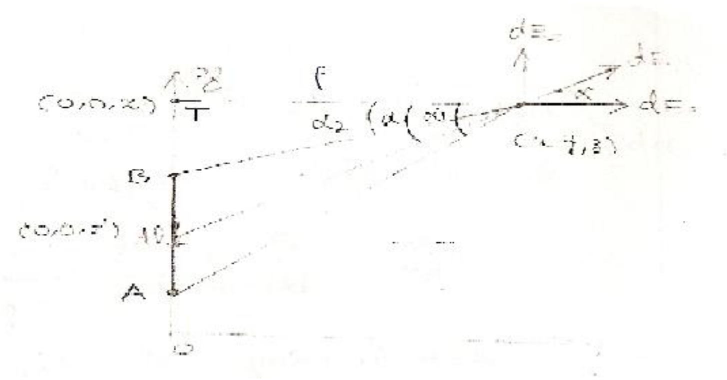
Line charge $E = \int_L \frac{\rho_L . dl}{4\pi \in_0 R^2} . i_R$

Surface charge $E = \int_S \frac{\rho_S . ds}{4\pi \in_0 R^2} . i_R$

Volume charge $E = \int_V \frac{\rho_V . dv}{4\pi \in_0 R^2} . i_R$

# -:Line charge:-

→ Consider a Line charge with uniform charge density $\rho_L$ extending from A to B along the z-axis as shown in fig. below



→ The elemental charge due to the elemental length dl = dz of the line is

$$dQ = \rho_L.dl = \rho_L.dz \qquad \rightarrow (1)$$

→ and hence the total charge Q is

$$Q = \int_{Z_A}^{Z_B} \rho_L.dz \qquad \rightarrow (2)$$

→ The electric field intensity E at an orbitary point P(x,y,z) can be found by

$$E = \int_l \frac{\rho_L.dl}{4\pi\in_0 .R^2} \qquad \rightarrow (3)$$

$dl = dz^1$

$\vec{R} = (x,y,z) - (0,0,z^1)$

$\vec{R} = x\, i_x + y\, i_y + (z-z^1)\, i_z$ (Rectangular)

(or)

$\vec{R} = \rho i_\rho + (z-z^1).i_z$ (Cylindrical)

$R^2 = |R|^2 = P^2 + (z-z^1)^2 = x^2+y^2+(z-z^1)^2$

$$\frac{iR}{R^2} = \frac{\vec{R}}{|R|^3} = \frac{\rho i_\rho + (z-z^1)i_z}{[\rho^2 + (z-z^1)^2]^{3/2}}$$

→ Substituting all this in equation (3)

$$E = \frac{\rho_L}{4\pi\in_0}\int \frac{\rho a_\rho + (z-z^1)i_z}{[\rho^2 + (z-z^1)^2]^{3/2}}.dz^1 \qquad \rightarrow (4)$$

→ To evaluate this, we define $\alpha$, $\alpha_1$ and $\alpha_2$ as shown in fig.

$|R| = [\rho^2 + (z - z^1)^2]^{1/2} = \rho \sec\alpha \quad \rightarrow (5)$

$z^1 = OT - \rho\tan\alpha$ $\qquad\qquad\qquad$ $z - z^1 = \rho\tan\alpha$

$dz^1 = -\rho.\sec^2\alpha.d\alpha \quad \rightarrow (6)$

→ Substituting these in equation (4)

$$E = \frac{\rho_L}{4\pi\in_0}\int_{\alpha_1}^{\alpha_2}\frac{[\rho i_\rho + \rho\tan\alpha\, i_z}{(\rho\sec\alpha)^3}x - \rho\sec^2\alpha.d\alpha$$

$$E = \frac{-\rho_L}{4\pi\in_0}\int_{\alpha_1}^{\alpha_2}\frac{i_\rho + \frac{\sin\alpha}{\cos\alpha}i_z}{\rho.\sec\alpha}.d\alpha$$

$$E = \frac{-\rho_L}{4\pi\in_0\rho}\int_{\alpha_1}^{\alpha_2}\frac{(\cos\alpha\, i_\rho + \sin\alpha\, i_z)}{\rho.\sec\alpha.\cos\alpha}.d\alpha$$

$$E = \frac{-\rho_L}{4\pi\in_0\rho}\left[\sin\alpha\, i_\rho + \cos\alpha\, i_z\right]_{\alpha_1}^{\alpha_2}$$

$$E = \frac{-\rho_L}{4\pi\in_0\rho}\left[(\sin\alpha_2 - \sin\alpha_1)i_\rho - (\cos\alpha_2 - \cos\alpha_1)i_z\right]$$

→ Thus for a finite Line charge

$$E = \frac{\rho_L}{4\pi\in_0\rho}[-(\sin\alpha_2 - \sin\alpha_1)i_\rho - (\cos\alpha_2 - \cos\alpha_1)i_z]$$

→ For an infinite Line charge, point B is at (0,0,-α) so that $\alpha_1 = \pi/2$, $\alpha_2 = -\pi/2$

$\because \alpha = 180^o,\ \alpha_1 - \alpha_2 = 180^o$

The z-component vanishes and

$$E = \frac{\rho_L.i_\rho}{2\pi\in_0.\rho}$$

Where $\rho$ the perpendicular distance from the Line to the point of is intersect, $i_\rho$ is the unit vector.

# -: Surface Charge:-

→ Consider an infinite sheet of charge in xy-plane with uniform charge density. The charge associated with an elemental area dS is

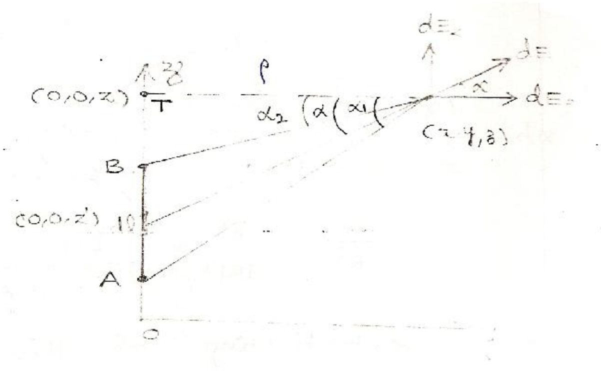
$$dQ = \rho_s.ds \quad \rightarrow (1)$$

→ and hence the total charge

$$Q = \int_s \rho_s.ds \quad \rightarrow (2)$$

→ The contribution to the electric field E at point P (0, 0, h) by the elemental surface shown in fig. is given by

$$dE = \frac{dQ}{4\pi \in_0 R^2}.i_R \quad \to (3)$$



→ From the fig.

$R = -\rho i_\rho + n i_z$

$|R| = (\rho^2 + n^2)^{1/2}$

$iR = \frac{\vec{R}}{|R|} = \frac{-\rho i_\rho + n i_z}{(\rho^2 + n^2)^{1/2}}$

$dQ = \rho_s ds = \rho_s.\rho d\phi d\rho$

→ Substituting these in eqn.(3) we get

$dE = \frac{\rho_s.\rho.d\phi d\rho[-\rho i_\rho + n i_z]}{4\pi \in_0 (\rho^2 + n^2)^{3/2}}$

→ Due to the symmentry lof lthe charge distribution, for every element 1, there is a corresponding element 2 whose contribution along $i_\rho$ cancels that of element 1. Thus the contributions to $E_\rho$ add up to zero.

So that E has only z-component.

→ $E = \int dE_z = \frac{\rho s}{4\pi \in_0} \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\alpha} \frac{h.\rho.d\phi.d\rho}{(\rho^2 + h^2)^{3/2}}.i_z$

→ $E = \frac{\rho_s.h}{4\pi \in_0} \int_{\phi=0}^{2\pi} d\phi.\frac{\rho}{(\rho^2 + h^2)^{3/2}}.d\rho.i_z$

→ $E = \frac{\rho_s.h}{4\pi \in_0} x 2\pi \int_{\rho=0}^{\alpha} d\phi.\frac{1}{(\rho^2 + h^2)^{3/2}}.d\rho.i_z$

→ $E = \frac{\rho s \, x \, h}{2 \in_0} \left[ -\frac{1}{\sqrt{\rho^2 + h^2}} \right] i_z$

→ $E = \frac{-\rho s h}{2 \in_0} \left[ 0 - \frac{1}{h} \right] i_z$

$$E = \frac{\rho_s}{2\epsilon_0} i_z$$

→ That is, E has only z-component if the charge is in the xy-plane.

→ In general, for an infinite sheet of charge,

$$E = \frac{\rho_s}{2\epsilon_0} . i_n$$

Where $i_n$ is a unit vector normal to the sheet.

→ The electric field intensity is independent of the distance b/w the sheet and the point of observation P.

→ In a parallel plate, capacitor, the electric field existing between the two plates having equal and opposite charges is given by

$$E = \frac{\rho_s}{2\epsilon_0} i_n + \frac{-\rho s}{2\epsilon_0}(-i_n) = \frac{\rho s}{\epsilon_0} i_n$$

**<u>Problem</u>**: A circular ring of radius 'a' carries a uniform charge $\rho_L\ c/m$ and is placed on the xy plane with axis the same as the z-axis.

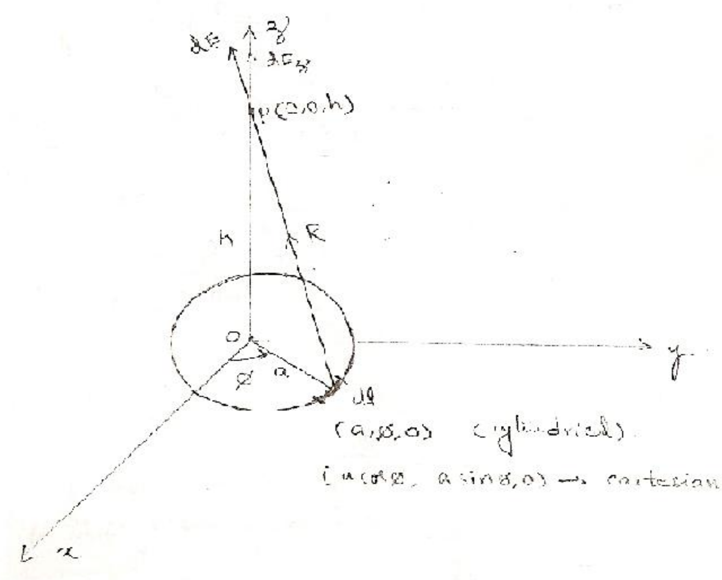
a) Show that $E(O,O,h) = \frac{\rho_L ah}{2\epsilon_0 [h^2 + a^2]^{3/2}} . i_z$

b) What values of 'h' gives the maximum value of E.

c) If the total charge on the ring is Q, find E as $a \to 0$.

<u>Solution:</u>

a)




→ $dl = a.d\phi$

$\vec{R} = (0,0,h) - (a\cos\phi, a\sin\phi, O)$

$\vec{R} = -ai_\rho + hi_z$

$|R| = [a^2+h^2]^{1/2}$

$$i_R = \frac{\vec{R}}{|\vec{R}|}$$

$$i_R = \frac{-ai_\rho + hi_z}{[a^2+h^2]^{½}}$$

$\rightarrow$Electric field intensity of any orbitary point P (0,0,h) is given by

$$E = \int_l \frac{\rho_l.dl}{4\pi\in_0 R^2}.i_R = \int_l \frac{\rho_l.dl}{4\pi\in_0 R^3}.\vec{R}$$

$$E = \frac{\rho_l}{4\pi\in_0}\int_{\phi=0}^{2\pi} \frac{(-ai_\rho + hi_z)}{[a^2+h^2]^{3/2}}.a.d\phi$$

$\rightarrow$By symmetry, the contributions along is add up to zero. i.e., for every element dl there is a corresponding element diametrically opposite, so that the two contributions cancel each other.

$\therefore$ The 'E' has only z-component.

$\rightarrow$That is

$$E = \frac{\rho_l}{4\pi\in_0}\int_{\phi=0}^{2\pi} \frac{h.a.d\phi}{[a^2+h^2]^{3/2}} i_z$$

$$E = \frac{\rho_l.h.a}{4\pi\in_0 [a^2+h^2]^{3/2}} \times 2\pi.i_z$$

$$E = \frac{\rho_l.h.a}{2\in_0 [a^2+h^2]^{3/2}} i_z$$

b)

$\rightarrow$Differentiating above equation with respect to 'n' and equation to zero for maximum condition of 'n'.

$$\frac{d|E|}{dh} = 0 \Rightarrow \frac{\rho_l.a}{2\in_0}\frac{[(a^2+h^2)^{3/2} \times 1 - \frac{3}{2} \times h(a^2+h^2)^{½} \times 2h]}{(a^2+h^2)^3} = 0$$

$\Rightarrow$ $(a^2+h^2)^{3/2} - 3h^2(a^2+h^2)^{½} = 0$

$(a^2+h^2)^{3/2} = 3h^2(a^2+h^2)^{½}$

$(a^2+h^2) = 3h^2$

$a^2 - 2h^2 = 0$

$a^2 = 2h^2$ (or) $h^2 = \frac{a^2}{2}$

$h = \pm\frac{a}{\sqrt{2}}$

d) Since the charge is uniformly distributed, the Line charge density is

$$\rho_L = \frac{Q}{2\pi a}$$

$$E = \frac{Q.a.h.i_z}{2\pi a.2\in_0.(h^2+a^2)^{3/2}}$$

$$E = \frac{Qh}{4\pi \in_0 .(h^2 + a^2)^{3/2}} .i_z$$

$\rightarrow$ As $a \rightarrow 0$

$$E = \frac{Qh}{4\pi \in_0 .h^3} .i_z$$

$$E = \frac{Q}{4\pi \in_0 .h^2} .i_z$$

(or) in general $E = \dfrac{Q}{4\pi \in_0 .R^2} .i_R$

Which is the same as that of a point charge.

**<u>Problem</u>**: A circular ring of radius 'a' carries uniform charge $\rho_L$ c/m and is in xy-plane. Find the electric field intensity at point (0,0,2) along its axis.

<u>Soln</u>:

$$E = \frac{\rho_L\, a.h}{2\pi \in_0 [a^2 + h^2]^{3/2}} .i_z$$

$$E = \frac{\rho_L\, a.2}{2\pi \in_0 [a^2 + 4]^{3/2}} .i_z$$

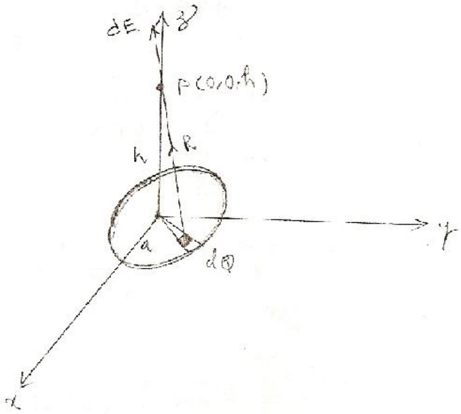
$$E = \frac{a.\rho_L}{\pi \in_0 [a^2 + 4]^{3/2}} .i_z$$

* * *

**Example**: A circular disk of radius 'a' is uniformly charged with $\rho_s$ $c/m^2$.
If the disk lies on the
z=0 plane with its axis along the z-axis.
a)Show that at point (0,0,h)

$$E = \frac{\rho_s}{2 \in_0} \left[ 1 - \frac{h}{(h^2 + a^2)^{1/2}} \right] i_z$$

b)From this, derive the E field due to an infinite sheet of charge on the z=0 plane.
c)If a<<h show that E is similar to the field due to a point charge.
Soln:




$$E = \frac{\rho_s ds}{4\pi \in_0 R^2}.i_R$$

$$\rightarrow dQ = \rho_s ds$$

$$dE = \frac{dQ}{4\pi \in_0 R^2}.i_R$$

$$dQ = \rho s.\rho.d\rho d\phi$$

$$\vec{R} = (0,0,h) - (\rho\cos\phi, \rho\sin\phi, 0)$$

$$\vec{R} = -\rho\cos\phi i_x - \rho\sin\phi i_y + hi_z$$

$$\vec{R} = \rho i_\rho + hi_z$$

$$\rightarrow |\vec{R}| = (\rho^2 + h^2)^{½}$$

$$\therefore\ dE = \frac{\rho s.\rho d\rho.d\phi(-\rho i_\rho + hi_z)}{4\pi \in_0 (\rho^2 + h^2)^{3/2}}$$

$$E = \frac{\rho_s}{4\pi \in_0} \int_{\rho=0}^{a} \int_{\phi=0}^{2\pi} \frac{\rho d\rho d\phi(-\rho i_\rho + hi_z)}{(\rho^2 + h^2)^{3/2}}$$

$\rightarrow$ due to symmetry, radial component vanishes

$$\therefore\quad E = \frac{\rho_s}{4\pi \in_0} \int_{\phi=0}^{2\pi} d\phi. \int_{\rho=0}^{a} \frac{\rho d\rho}{(\rho^2 + h^2)^{3/2}}.hi_z$$

$$E = \frac{\rho_s}{4\pi \in_0}.2\pi.h\left[\frac{-1}{(\rho^2 + h^2)^{3/2}}\right]_{\rho=0}^{a} i_z$$

$$E = \frac{-\rho_s}{2 \in_0}.h\left[\frac{1}{\sqrt{h^2 + a^2}} - \frac{1}{h}\right] i_z$$

$$E = \frac{-\rho_s}{2 \in_0}\left[\frac{h}{\sqrt{h^2 + a^2}} - 1\right] i_z$$

$$E = \frac{\rho_s}{2 \in_0}\left[1 - \frac{h}{(a^2 + h^2)^{½}}\right] i_z$$

* * *

**Example**: A uniform Line charge, infinite in extent with $\rho_l = 20 nc/m$, lies along the z-axis. Find E at (6,8,3) m.

Soln:

$\rightarrow$Electric field intensity due to the infinite Line charge is given by

$$E = \frac{\rho_l}{2\pi \in_0 \rho} . i_\rho$$

$$\rho = \sqrt{(6)^2 + (8)^2} = \sqrt{36+64} = \sqrt{100} = 10m$$

Which is the perpendicular distance from (6, 8, 3) to the z-axis.

$\rightarrow$ Given $\rho_l = 20x10^{-9} c/m$

$$\therefore \quad E = \frac{20x10^{-9}}{2\pi x \frac{1}{36\pi} x10^{-9} x10} x i_\rho$$

$$E = 36\ i_\rho\ v/m$$

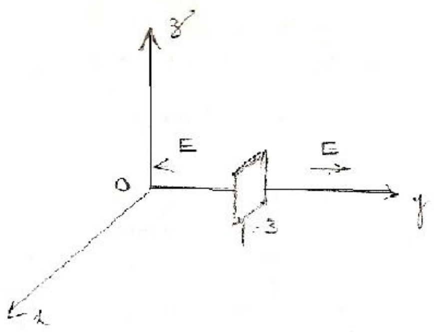
**Example**: Charge is distributed uniformly over the plane z = 10cm with a density $\rho s = \frac{1}{3\pi} nc/m^2$. Find E above the sheet. (z > 10cm) & below (z < 10cm) the sheet.

Sol: Above the sheet z > 10cm $\quad E = \frac{\frac{1}{3\pi} x10^{-9}}{2x \frac{1}{36\pi} x10^{-9}} i_z = 6\ i_z$ v/m

Below the sheet z < 10cm $\quad E = -6\ i_z$ v/m

**Example**: The plane y = 3m contains a uniform charge distribution of density $\rho_s = \frac{10^{-8}}{6\pi} c/m^2$. Determine E at all points.

Given $\rho s = \frac{10-8}{6\pi} c/m^2$




Soln: For y > 3mt

$$E = \frac{\rho_s}{2\in_0}.i_y$$

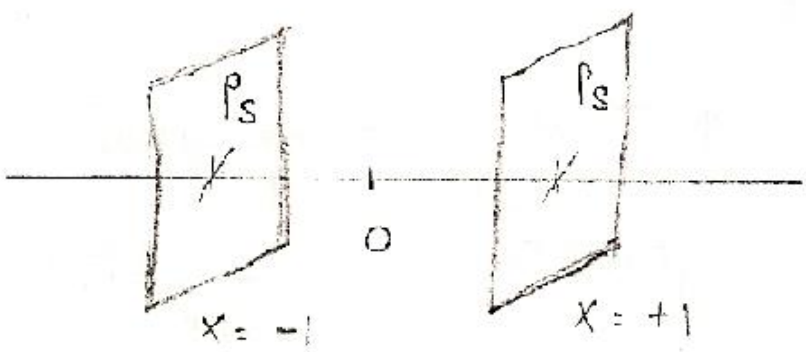
$$E = \frac{10^{-8}}{6\pi \, x \, 2 \, x \, \frac{1}{36\pi} x 10^{-9}}.i_y$$

$E = 30\ i_y$ v/m

For y < 3mt    $E = -30\ i_y$ v/m

**Example**: Two infinite uniform sheets of charge, each with density $\rho_s$ are located at x = ±1 as shown in figure. Determine E in all regions.

Soln:




i) x < -1

$E = -\frac{\rho_s}{\in_0} i_x$

ii) -1 < x < 1    E = 0

iii) x > 1    $E = \frac{\rho_s}{\in_0} i_x$

Repeat the above problem with

x = -1 and $-\rho_s$ on x = ±1

Soln:

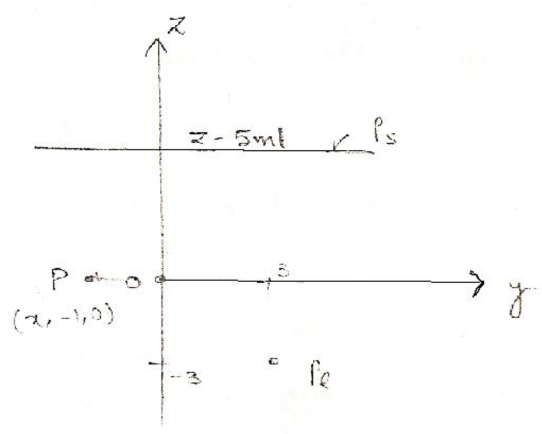
i) x < -1    E = 0

ii) -1 < x < 1    $E = \frac{\rho_s}{\in_0} i_x$

iii) x > 1    E = 0

Example: A uniform sheet charge with $\rho s = \frac{1}{3\pi} nc/m^2$ is located at z = 5m and a uniform Line charge with $\rho_l = -\frac{25}{9} nc/m$ at z = -3m, y = 3m. Find E at (x,-1, 0) m.

Soln:




Given

$$\rho_s = \frac{1}{3\pi} x10^{-9}\ c/m^2$$

$$\rho_l = -\frac{25}{9} x10^{-9}\ c/m$$

$\rightarrow$ Due to the sheet charge $E_s = \frac{\rho_s}{2\in_0}(-i_n)$

$$E_s = \frac{1}{3\pi} x \frac{10^{-9}}{2x\frac{1}{36\pi}x10^{-9}} x(-i_z)$$

$$E_s = -6i_z \text{ v/m}$$

$\rightarrow$ Due to the Line charge

$$E_l = \frac{E_l}{2\pi\in_0\rho}.i_\rho$$

(or)

$$E_l = \frac{\rho_l}{2\pi\in_0 r}.i_r$$

$$E_l = \frac{-\frac{25}{9}x10^{-9}}{2\pi x\frac{1}{36\pi}x10^{-9}} \frac{[-4i_y + 3i_z]}{25}$$

$$E_l = +8i_y - 6i_z \text{ v/m}$$

$\rightarrow$ $\therefore$ E at (x,-1, 0) m due to sheet charge and Line charge is

$$E = E_s + E_l$$

$$E = -6i_z + 8i_y - 6i_z$$

$$E = 8i_y - 12i_z \text{ v/m}$$

***

**Example**: Two uniform Line charges of density $\rho_l = 4\,\text{nc/m}$ lie in the x = 0 plane at y = ±4m. Find E at (4, 0, 10) m.

Soln: $E = \frac{\rho_l}{2\pi \in_0} \frac{[4i_x - 4i_y]}{[32]} + \frac{\rho_l}{2\pi \in_0} \frac{[4i_x + 4i_y]}{[32]}$

$$E = \frac{\rho_l}{\pi \in_0} \frac{4i_x}{32}$$

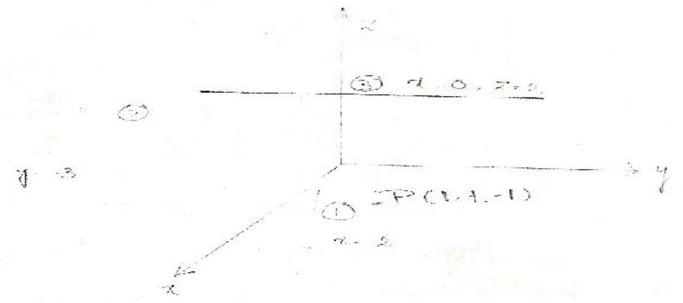
$$E = \frac{4x10^{-9} x4}{\pi \frac{1}{36\pi} x10^{-9} x32} .i_x \ = 18\ i_x \text{ v/m}$$

***

**Example**: Planes x = 2 and y = -3 respectively, carry charges 10 nc/m² and 15 nc/m². If the Line x = 0, z = 2 carries charge $10\pi$ nc/m, calculate E at (1,1,-1) due to the three charge distributions.

Ans:- Let $E = E_1 + E_2 + E_3$

Where $E_1$, $E_2$ and $E_3$ are respectively, the contributions to E at point (1,1,-1) due to infinite sheet 1, infinite sheet 2, and infinite Line 3 as shown in fig.




$$E_1 = \frac{\rho_s}{2 \in_0} (-i_x)$$

$$E_1 = \frac{10x10^{-9}}{2x \frac{1}{36\pi} x10^{-9}} (-i_x)$$

$$E_1 = -180\pi\ i_x \quad \rightarrow (1)$$

$$E2 = \frac{\rho_s}{2 \in_0} (i_y) = \frac{-15x10^{-9}}{2x \frac{1}{36\pi} x10^{-9}} (i_y)$$

$$E2 = 270\pi i_y \quad \rightarrow (2)$$

$\rightarrow$ $$E3 = \frac{\rho_l}{2\pi \in_0 \rho} .i_\rho$$

Where $\rho$ is the perpendicular distance from point P(1,-1,1) to the Line.

x = 0, z=2

[0,y,2] [1,1,-1]

$\vec{\rho} = i_x - 3i_z$

$|\rho| = \sqrt{1+9} = \sqrt{10}$

$i_\rho = \dfrac{i_x - 3i_z}{\sqrt{10}}$

$E_3 = \dfrac{10\pi \, x 10^{-9}}{2\pi \, x \dfrac{1}{36\pi} x 10^{-9}} x \dfrac{i_x - 3i_z}{10}$

$E_3 = 18\pi(i_x - 3i_z) \quad \rightarrow (3)$

$\rightarrow$ Thus by adding E1, E2 and E3 we get the total field $E = E_1 + E_2 + E_3$

$E = -180\pi i_x + 270\pi i_y + 18\pi(i_x - 3i_z)$

$E = -162\pi i_x + 270\pi i_y - 54\pi i_z \ \ v/m$

***

# -: <u>Electric Flux($\psi$)</u>:-
# (OR)
# -:<u>Electric Displacement</u>:-

$\rightarrow$**Faraday's experiment:-**

A sphere with charge 'Q' was placed with but not touching, a larger hollow sphere. The outer sphere was earthed momentarily, and then the inner sphere was removed.

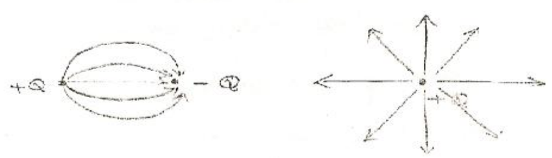
The charge remaining on the outer sphere was then measured. This charge was found to be equal (and of opposite sign) to the charge on the inner sphere for all sizes of the spheres and for all types of dielectric media b/w the spheres. This could be considered that there was on electric displacement from the charge on the sphere through the medium to the outer sphere. The amount of this displacement depending only upon the magnitude of the charge Q.

i.e. $\psi = Q$

$\rightarrow$Electric Flux $(\psi)$ is a scalar field.

$\rightarrow$Electric Flux originates on positive charge and terminates on Negative charge.

$\rightarrow$In the absence of Negative charge, the flux $(\psi)$ terminates at infinity.



→By definition, one coulomb of electric charge gives rise to one coulomb of electric flux.

## ELECTRIC FLUX DENSITY:-

→Electric flux density is defined as the ratio of the electric flux per unit area.
→Electric flux density 'D' is a vector field and is measured in coulombs per square meter.
→Electric flux density 'D' at any point on a spherical surface of radius 'r' centered at the isolated charge 'Q' is given by

$$D = \frac{\psi}{4\pi r^2} = \frac{Q}{4\pi r^2} \ c/m^2$$

→Electric flux density is in the radial direction i.e., normal to the surface.

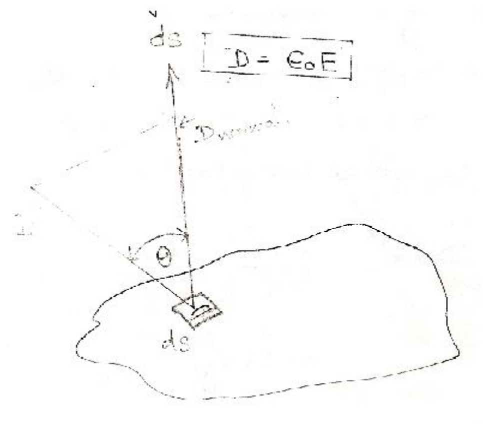
$$D = \frac{Q}{4\pi r^2} i\hat{r} \ c/m^2 \quad \rightarrow (1)$$

→The electric field intensity at any point due to a point charge is given by

$$E = \frac{Q}{4\pi \in_0 R^2} i_R \quad \rightarrow (2)$$

i.e., at any particular point the electric field strength E depends not only lupon the magnitude and position of the charge Q, but also upon the di-electric constant of the medium in which the field is measured.
→The electric flux density D is independent of the medium.
→Comparing equations (1) & (2) we get

$$D = \in_0 E$$




→The flux crossing through an elemental surface ds is the product of the normal component of D and ds.

$$d\psi = D\cos\theta.ds$$
$$d\psi = Dds\cos\theta$$
$$d\psi = D.ds$$

$\rightarrow$The total flux passing through the closed surface is obtained by adding the different contributions crossing each surface element ds.

$$\psi = \int d\psi = \oint_S D.ds$$

$\rightarrow$The electric flux density 'D' due to point charge:

$$D_Q = \frac{Q}{4\pi R^2}.i_R$$

$\rightarrow$Due to Line charge: $D_l = \frac{\rho_L}{2\pi\rho}.i_\rho$

$\rightarrow$Due to an infinite sheet of charge.

$$D = \frac{\rho_S}{2} i_n$$

**Example**: Determine D at (4,0,3) if there is a point charge -5π mc at (4,0,0) and a Line charge 3π mc/m along the y-axis.

Soln:

$$D_Q = \frac{-5\pi \times 10^{-3}}{4\pi} \times \frac{(3i_z)}{[3]^3}$$

$$D_Q = -0.138\ i_z\ \ mc/m^2$$

$$D_l = \frac{3\pi \times 10^{-3}}{2\pi} \frac{(4i_x + 3i_z)}{[25]}$$

$$D_l = 0.24\ i_x + 0.18\ i_z\ \ mc/m^2$$

$$D = D_Q + D_l$$

$$D = 240ix + 42\ iz\ \ \mu c/m^2$$

***

# -: Gauss's Law:-

"Gauss's Law states that the total electric flux out of a closed surface is equal to the total charge enclosed by that surface".

Thus $\psi = Q_{enclosed}$

$\rightarrow$This can be written in integral form as

$$\psi = \int d\psi = \oint_S D.ds = Q_{enclosed} \quad \rightarrow (2)$$

$\rightarrow$The total charge enclosed

$$Q = \int_V \rho_V dv \quad \rightarrow (3)$$

$\rho_v \rightarrow$ Volume charge density

$\rightarrow$Equating equation (2) & (3)

$Q = \oint_S D.ds = \int_V \rho_V dv \rightarrow (4) \rightarrow$ Integral form of +1 gauss law.

$\rightarrow$By applying the divergence theorem to the middle term of the above equation

$$\oint_S D.ds = \int_V \nabla.D\,dv \rightarrow (5)$$

$\rightarrow$Comparing equations (4) & (5)

$$\int_V \nabla.D\,dv = \int_V \rho_V dv$$

$$\nabla.D = \rho_V$$

$\rightarrow$ Differential (or) point from of the gauss law which is the Maxwell's first equation.

$\rightarrow$ Gauss's Law provides on easy means of finding E or D for symmetrical charge distributions.

**APPLICATIONS OF GAUSS'S LAW:-**

$\rightarrow$ Conditions (or) procedure for applying Gauss's Law

i) Symmetric charge distribution.

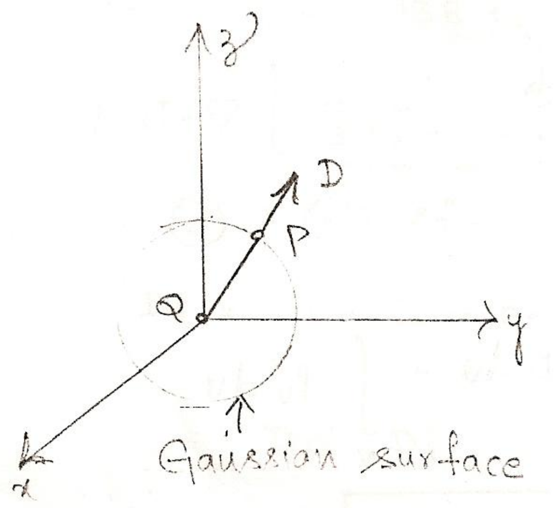
ii) Closed surface [Gaussian surface]

iii) D is everywhere either normal (or) tangential to the closed lsurface, so that D.ds becomes either Dds (or) zero respectively.

# A.Point Charge:-

$\rightarrow$ Suppose a point charge Q is located at the origin. To determine D at a point P, it is easy lto see that choosijng a spherical surface containing P will satisfy symmetry conditions.

$\rightarrow$ i.e., a spherical surface centered at the origin is the Gaussian surface.




$\rightarrow$ Since D is every where normal to the Gaussian surfaces that is D = $D_r i_r$

$\rightarrow$ Applying the Gauss's Law

$Q = \oint D.ds = D_r \oint ds$

$Q = Dr\, 4\pi\, r^2$

Where $\oint ds = \int_{\phi=0}^{2\pi} d\phi . \int_{\theta=0}^{\pi} r^2 \sin\theta\, d\theta$

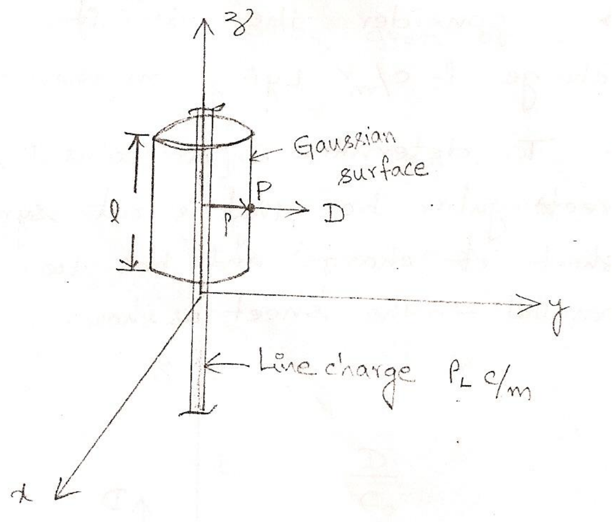
$\oint ds = 4\pi r^2$

$\therefore\ D = \dfrac{Q}{4\pi r^2} . i_r$

# **B.Infinite Line charge:-**

→ Suppose the infinite line of uniform charge $\rho_L$ c/m lies along the z-axis.

→ To determine D at a point P, we choose a cylilndrical surface containing P to satisfy symmetric condition.
→ D is constant on and normal to the cylindrical Gaussian surface. i.e., D $= D_\rho i_\rho$




→By applying Gauss's Law to an arbitrary length 'l' of the line
$Q = \oint_s D.ds = D\rho \oint_s ds$

$$= D_\rho \int_{\phi=0}^{2\pi} \rho\, d\phi . \int_{z=0}^{l} dz$$

$$= \rho D_\rho .2\pi .l$$

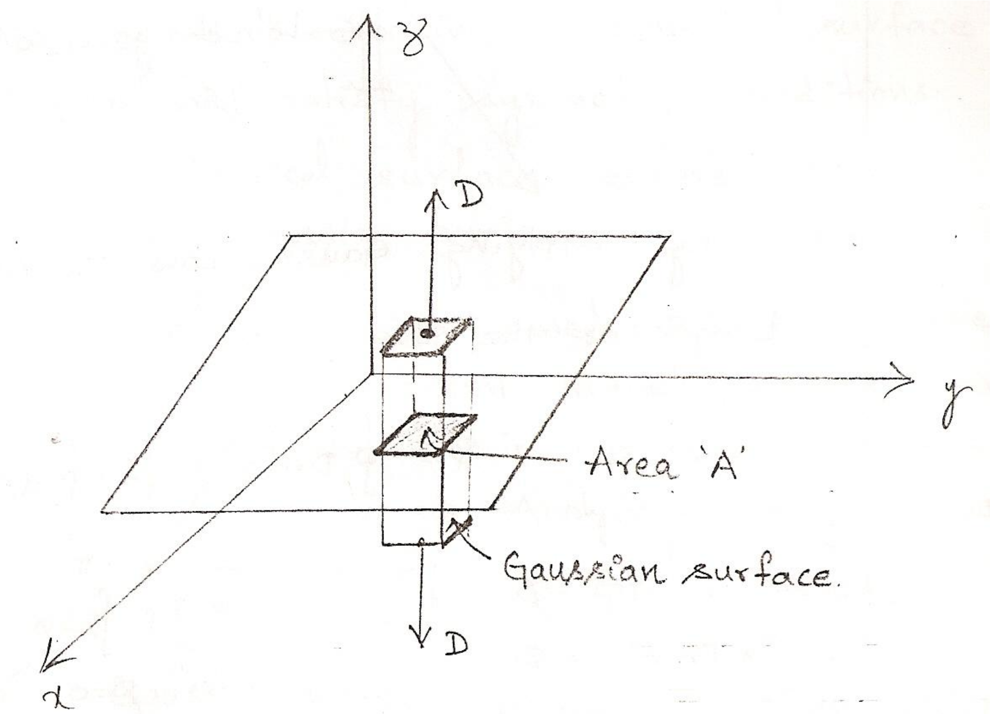
$$Q = \rho_l .l = D_\rho .2\pi\rho .l$$

$$D = \frac{\rho_L}{2\pi\rho} . i_\rho$$

$\int D.ds \rightarrow \int D.ds$ evaluated on the top and bottom surfaces of the cylinder is zero since D has no z-component, i.e., D is tangential to those surfaces.

## C.Infinite sheet of charge:-

$\rightarrow$Consider the infinite sheet of uniform charge $\rho_S \ c/m^2$ lying on the z = 0 plane.
$\rightarrow$To determine D at point P, we choose a rectangular box that is cut symmetrically by the sheet of charge land has two of its faces parallel to the sheet as shown.




$\rightarrow$As D is normal to the sheet, D = $D_z$iz
$\rightarrow$By applying gauss's law gives

$$Q = \oint_S D.ds \qquad = D_Z \oint_S ds$$

$$= D_Z \left[ \int_{top} ds + \int_{bottom} ds \right]$$

$$= D_Z[A+A]$$

$$Q \qquad = 2AD_Z$$

$\rightarrow$If the top and bottom area of the box each has area A.

$$Q = \int \rho_S ds = 2AD_Z$$

$$= \rho_S.A = 2AD_Z$$

$$D = \frac{\rho_S}{2} i_z$$

$$\therefore \; E = \frac{D}{\epsilon_0}$$

$$E = \frac{\rho_S}{2\epsilon_0} i_z$$

***

# <u>D.Uniformly charged sphere</u>:-

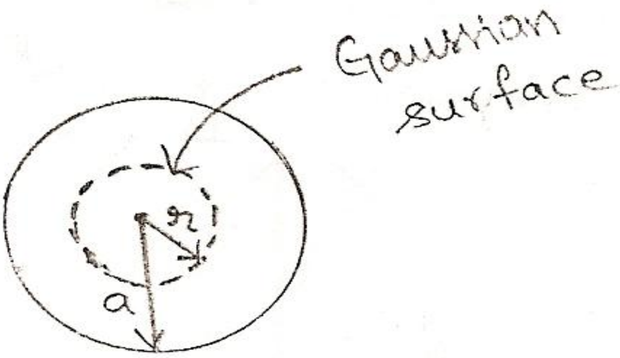
→Consider a sphere of radius ‘a’ with uniform charge $\rho_v \, c/m^3$.

→To determine ‘D’ every where we construct Gaussian surfaces for cases $r \le a$ and $r \ge a$ separately.

→Since the charge has spherical symmetry it is obvious that a spherical surface is an appropriate Gaussian surface.

**Case i)**

→For $r \le a$




The charge enclosed by the gaussian surface is

$$Q_{enclosed} = \int \rho v \, dv$$

$$= \rho_v \int_v dv$$

$$= \rho_v \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r} r^2 \sin\theta \, d\theta \, d\phi \, dr$$

$$= \rho_v \int_{r=0}^{r} r^2 dr \int_{\theta=0}^{\pi} \sin\theta \, d\theta \int_{\phi=0}^{2\pi} d\phi$$

$\rightarrow$ $Q_{enclosed} = \rho_v.\frac{r^3}{3}.2.2\pi$

$$Qenclosed = \rho_v.\frac{4}{3}.\pi r^3 \qquad \rightarrow(1)$$

$\rightarrow$ And

$$\psi = \oint_s D.ds$$

$$= D_r.\oint_s ds$$

$$= D_r.\oint_S D.ds$$

$$= D_r.\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} r^2 \sin\theta\, d\theta\, d\phi$$

$$= D_r.r^2.\int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin\theta\, d\theta$$

$$= Dr.r^2.2\pi x2$$

$$\psi = 4\pi r^2 D_r \qquad \rightarrow(2)$$

$\rightarrow$ By applying Gauss's law

$$\psi = Q_{enclosed}$$

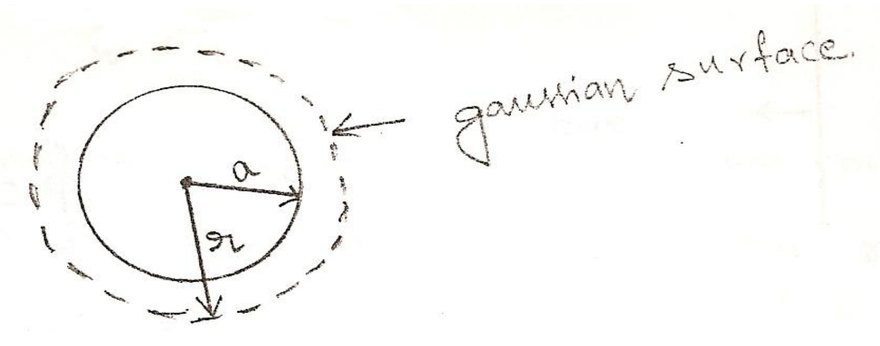
$$\therefore \quad 4pr^2\, D_r = \rho_v.\frac{4}{3}\pi r^3$$

$$D_r = \frac{r.\rho_v}{3}$$

$$\therefore \quad D = \frac{r}{3}.\rho_v.i_r \qquad o < r \le a$$

**Case (ii)**

$\rightarrow$For $r \ge a$




$\rightarrow$The charge enclosed by the Gaussian surface is

$$Qenclosed = \int_v \rho_v dv$$

$$= \rho_v \int_v dv = \rho v \int_v r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= \rho_v \int_{r=0}^{a} r^2 dr \int_{\theta=0}^{\pi} \sin\theta\, d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$\text{Qenclosed} = \rho_v . \frac{a^3}{3} .2.2\pi = \frac{4}{3}\pi a^3 . \rho_v \rightarrow (1)$$

$\rightarrow$While $\psi = \oint_s D.ds = Dr \oint_s ds$

$$= D_r r^2 \int_{\theta=0}^{\pi} \sin\theta\, d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$\psi = Dr.r^2.2.2\pi = 4\pi r^2 D_r \rightarrow (2)$$

$\rightarrow$Applying Gauss Law

Qenclosed = $\psi$

$$4/3\pi a^3 \rho_v = 4\pi r^2 D_r$$

$$D_r = \frac{a^3}{3r^2} . \rho_v$$

$$D = \frac{a^3}{3r^2} . \rho_v . i_r \quad \text{for } r \geq a$$

# Electric Potential

- $\rightarrow$ The amount of work done in moving a unit positive charge 'Q' from infinity to the required point is called "Potential".
- $\rightarrow$ Electric Potential is a scalar quantity.
- $\rightarrow$ To move a point charge 'Q' from point A to point B in an electric field 'E' the force on a point charge 'Q' is given by

$$F = EQ \rightarrow (1)$$

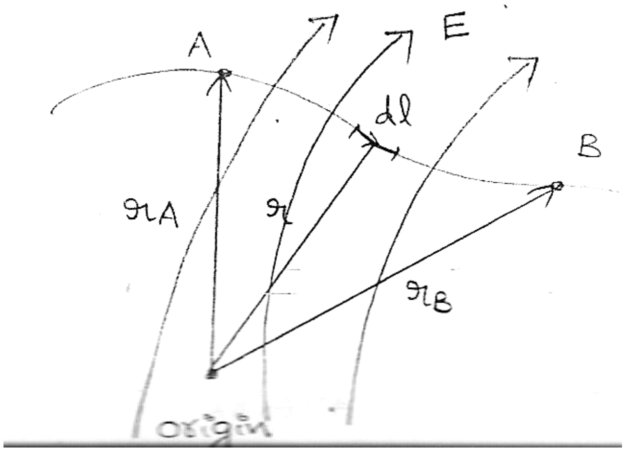
- $\rightarrow$ The amount of work done in displacing the charge by dl is

$$dW = - F.dl$$

$$dW = - EQ.dl \rightarrow (2)$$

- $\rightarrow$ The negative sign indicates that the work is being done by an external agent.
- $\rightarrow$ $\therefore$ Thus the total work done (or) the potential energy required, in moving 'Q' from A to B is

$$w = -Q\int_A^B E.dl \rightarrow (3)$$




→ Dividing W by Q gives the potential energy per unit charge, and is denoted by $V_{AB}$ which is known as the potential difference between points A & B.

$$\text{Thus } V_{AB} = \frac{W}{Q} = -\int E.dl \quad \rightarrow (4)$$

→ VAB is independent of path taken.

→ VAB is measured in Jouls/coulomb (or) volts.

→ The electric field intensity 'E' due to a point charge Q located at the origin is given by

$$E = \frac{Q}{4\pi \in_0 r^2}.i_r \quad \rightarrow (5)$$

→ Substituting eqn.(5) in (4)

$$V_{AB} = -\int_{r_A}^{r_B} \frac{Q}{4\pi \in_0 r^2}.i_r.dr.i_r$$

$$V_{AB} = -\frac{Q}{4\pi \in_0} \int_{r_A}^{r_B} \frac{1}{r^2} dr$$

$$V_{AB} = \frac{Q}{4\pi \in_0} \left[\frac{1}{r}\right]_{r_A}^{r_B}$$

$$V_{AB} = \frac{Q}{4\pi \in_0} \left[\frac{1}{r_B} - \frac{1}{r_A}\right]_{r_A}^{r_B}$$

$$V_{AB} = \frac{Q}{4\pi \in_0 r_B} - \frac{Q}{4\pi \in_0 r_A}$$

$$V_{AB} = V_B - V_A$$

→ Where VB & VA are the potentials (or absolute potentials) at B and A respectively.

→ If we assume that potential at infinity is zero i.e., $V_A$=0 as $r_A \rightarrow \infty$ Then the potential at any point due to the point charge 'Q' located at the origin is

$$V = -\int_{\infty}^{r} E.dl$$

→ If the point charge 'Q' is not located at the origin but at a point whose position vector is $r^1$. Then the potential at r becomes

$$V(r) = \frac{Q}{4\pi \in 0\,|r - r^1|}$$

→ For n point charges $Q_1, Q_2, \ldots . Q_n$ located at points with position vector's $r_1, r_2, r_3, \ldots .. r_n$ the potential at r is

$$V(r) = \frac{Q_1}{4\pi \in_0 |r - r_1|} + \frac{Q_2}{4\pi \in_0 |r - r_2|} + \ldots\ldots\ldots + \frac{Q_n}{4\pi \in_0 |r - r_n|}$$

(or)

$$V(r) = \frac{1}{4\pi \in_0} \sum_{K=1}^{n} \frac{Q_K}{|r - r_K|}$$

→ For continuous charge distributions, we replace $Q_K$ in above eqn. with charge element $\rho_L dl$, $\rho_s ds$ (or) $\rho_v dv$ and the summation becomes integration.

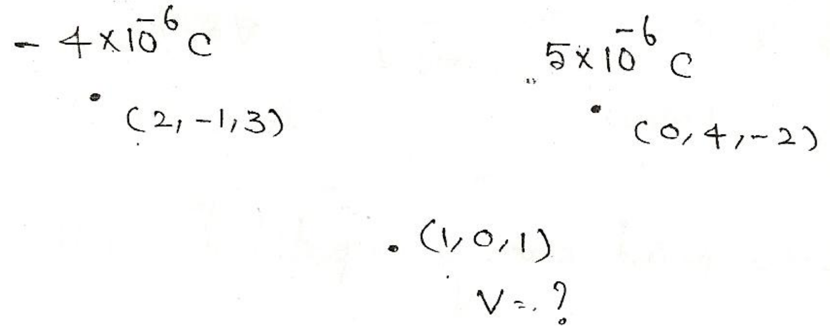
$\therefore$ The potential at 'r' becomes

$$V(r) = \frac{1}{4\pi \in_0} \int_l \frac{\rho_L\, dl}{|r - r^1|} \qquad \text{(Line charge)}$$

$$V(r) = \frac{1}{4\pi \in_0} \int_S \frac{\rho_S . ds}{|r - r^1|} \qquad \text{(Surface charge)}$$

$$V(r) = \frac{1}{4\pi \in_0} \int_V \frac{\rho_V . dv}{|r - r^1|} \qquad \text{(Volume charge)}$$

**Ex: 1**:Two point charges $-4\mu C$ and $5\mu C$ are located at (2,-1,3) and (0,4,-2) respectively. Find the potential at (1,0,1) assuming zero potential at infinity.




Given zero potential at infinity.

$$V = \frac{-4\,x\,10^{-6}}{4\pi\, x \frac{1}{36\pi} x 10^{-9}}\, x \frac{1}{|(1,0,1) - (2,-1,3)|} + \frac{5\,x\,10^{-6}}{4\pi\, x \frac{10^{-9}}{36\pi}}\, x \frac{1}{|(1,0,1) - (0,4,-2)|}$$

$$V = 9 x 10^3 \left[ \frac{-4}{\sqrt{1+1+4}} + \frac{5}{\sqrt{1+16+9}} \right]$$

$$V = 9 \text{ x } 10^3 \left( \frac{-4}{\sqrt{6}} + \frac{5}{\sqrt{26}} \right)$$

V = 9 x [$10^3$-1.633+0.9806]

V = -5.872 KV

**Ex: 2**: A point charge 5 nc is located at (-3,4,0) while line y = 1, z = 1 carries uniform charge 2 nc/m

a) If V = 0V at O(0,0,0), find V at A(5,0,1)

b) If V = 100V at B(1,2,1), find V at C(-2,5,3)

c) If V = -5V at O(0,0,0), find $V_{BC}$

Soln: Given 5nCis located at (-3, 4, 0)

Line is y = 1, z = 1 (x, 1, 1)

And $\rho_l = 2 \text{ x } 10^{-9}$ c/m

a) V = 0 (0, 0, 0)      V =? A = (5,0,1)

Due to point charge potential diffence

$$V_{OA} = V_A - V_O = \frac{Q}{4\pi \in_0} \left[ \frac{1}{r_A} - \frac{1}{r_O} \right]$$

$r_A = |(5,0,1) - (-3,4,0)|$

$r_A = |(8,-4,1)| = \sqrt{64+16+1} = \sqrt{81} = 9$

$r_0 = |(0,0,0) - (-3,4,0)|$

$r_0 = |(3,-4,0)| = \sqrt{9+16} = \sqrt{25} = 5$

$$V_A - V_0 = \frac{5x10^{-9}}{4\pi \, x \, \frac{10^{-9}}{36\pi}} \left[ \frac{1}{9} - \frac{1}{5} \right]$$

$$V_A - 0 = 45 \left[ \frac{5-9}{45} \right]$$

$V_A$ = -4 Volts

$\rightarrow$For the infinite Line charge

$$V_{OA} = -\int E.dl$$

$$V_{OA} = -\frac{\rho_l}{2\pi \in_0} i_\rho . d\rho . i_\rho$$

$$V_{OA} = -\frac{\rho_l}{2\pi \in_0} [\ln \rho]_0^A$$

$$V_{OA} = -\frac{\rho_l}{2\pi\in_0}\left[\ln\frac{\rho_A}{\rho_O}\right] = \frac{\rho_l}{2\pi\in_0}\left[\ln\frac{\rho_0}{\rho_A}\right]$$

$$V_A\text{-}V_0 = \frac{\rho_l}{2\pi\in_0}\left[\ln\frac{\rho_0}{\rho_A}\right]$$

$$\rho_A = |(5,0,1)-(x,1,1)| = |(0,-1,0)| = 1$$

$$\rho_B = |(0,0,0)-(x,1,1)| = |(0,-1,-1)| = \sqrt{2}$$

$$V_{A\text{-}0} = \frac{2\,x\,10^{-9}}{2\pi\,x\,\frac{10^{-9}}{36\pi}}\left[\ln\left[\frac{\sqrt{2}}{1}\right]\right]$$

$$V_A = 36[\ln\sqrt{2}\,]$$

$\therefore$ The potential at point A due to point charge and Line charge having v=0 at origin is given by

$$V = 36(\ln\sqrt{2}\,) - 4$$

$$V = 8.477 \text{ Volts}$$

**b)** 5 x 10-9 (x,1,1)

(-3,4,0) $\rho_l = 2\,x\,10^{-9}$ c/m

V = 100 V V =?

B = (1,2,1) C = (-2,5,3)

$\rightarrow$ Due to point charge

$$V_{BC} = V_C\text{-}V_B = \frac{5x10^{-9}}{4\pi\,x\,\frac{10^{-9}}{36\pi}}\left[\frac{1}{|(-2,5,3)-(-3,4,0)|} - \frac{1}{|(1,2,1)-(-3,4,0)|}\right]$$

$$V_{BC} = V_C\text{-}100V = 45\left[\frac{1}{|(1,1,3)} - \frac{1}{(4,-2,1)}\right]$$

$$V_{BC} = V_C\text{-}100V = 45\left[\frac{1}{\sqrt{1+1+9}} - \frac{1}{\sqrt{16+4+1}}\right]$$

$$V_{BC} = V_C\text{-}100V = 45\left[\frac{1}{\sqrt{11}} - \frac{1}{\sqrt{21}}\right]$$

$\Rightarrow$ $V_C$ = -100V = 45[0.08329]

$V_C$ = 100 + 3.748 $V_{BC}$ = 3.748

$V_C$ = 103.748 volts.

$\rightarrow$Due to the Line charge

$$V_{BC} = \frac{\rho_l}{2\pi\in_0}\left[\ln\frac{\rho_B}{\rho_C}\right] = 36 \ln \frac{1}{\sqrt{20}}$$

$\rightarrow$Due to both point charge & Line charge

$$V_{BC} = 3.748 + 36 \ln \frac{1}{\sqrt{20}}$$

$$V_{BC} = -50.175$$

$$V_C\text{-}V_B = -50.175$$

$V_C = 49.82V$

**Example:** A Line charge $\rho_L = 400\rho c/m$ lies along the x-axis. The surface of zero potential passes through the point (0, 5, 12) m. Find the potential at point (2,3,-4)m.

Soln: Given rL=400x10-12 c/m lies along the x-axis i.e., (x, 0, 0)

Let V=0 V=?

A (0, 5, -12) m B (2, 3, -4)

$\rightarrow$The potential due to the Line charge

$$V_{AB} = - \frac{\rho_l \, d\rho}{2\pi \in_0 \rho} \text{ , Where } \rho$$

$$V_{AB} = -\frac{\rho_l}{2\pi \in_0}[\ln \rho]_A^B$$

$$V_{AB} = -\frac{\rho_l}{2\pi \in_0}[\ln \rho_B - \ln \rho_A]$$

$$V_{AB} = -\frac{\rho_l}{2\pi \in_0}\left[\ln \frac{\rho_A}{\rho_B}\right]$$

$$\rho_A = |(0,5,-12)-(x,0,0)| = |(0,5,12)|$$

$$\rho_A = \sqrt{25+144} = \sqrt{169} = 13m$$

$$\rho_B = |(2,3,-4)-(x,0,0)| = |(0,3,-4)|$$

$$\rho_B = \sqrt{9+16} = \sqrt{25} = 5m$$

$$V_{AB} = \frac{400x10^{-12}}{2\pi x \frac{10^{-9}}{36\pi}}\left[\ln\left(\frac{13}{5}\right)\right] = 18x4x10^{-1}\ln\left[\frac{13}{5}\right] = 6.879V$$

$V_B - V_A = 6.879V$

$V_B = 6.879V$

**Ex:** A point charge of 15nc is situated at the origin and another point charge of -12nc is located at the point (3, 3, 3) m. Find E and V at the point (0,-3,-3).

Soln: 15x10-9C -12x10-9

. .

(0, 0, 0) (3, 3, 3)

E =? V =?

. 0,-3,-3)

$$E = \frac{15X10^{-9}}{4\pi \in_0 \left(\sqrt{18}\right)^3}(-3i_y - 3i_z) + \frac{-12x10^{-9}}{4\pi \in_0}\frac{(-3ix - 6iy - 6iz)}{\left[\sqrt{9+36+36}\right]^3}$$

$$E = \frac{-3x15x9(i_y + i_z)}{\left[\sqrt{18}\right]^3} + \frac{12x9x3\left[i_x + 2i_y + 2i_z\right]}{\left[\sqrt{9+36+36}\right]^3}$$

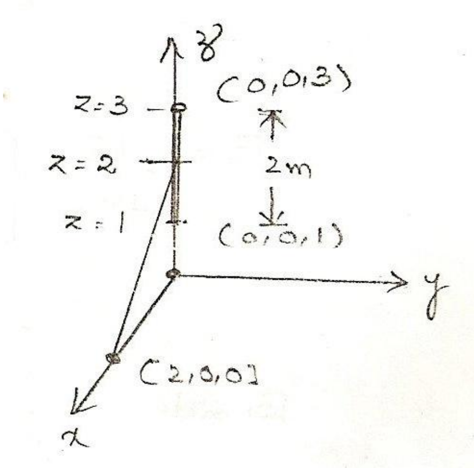
$E = -5.3033[i_y+i_z] + 0.444[i_x+2i_y+2i_z]$

$E = 0.444 i_x - 4.414 i_y - 4.414 i_z$

$$V = \frac{15 x 10^{-9}}{4\pi x \frac{10^{-9}}{36\pi} |(0,-3,-3)} + \frac{-12 x 10^{-9}}{4\pi x \frac{10^{-9}}{36\pi} |(-3,-6,-6)|}$$

$$V = \frac{15 x 9}{\sqrt{9+9}} - \frac{12 x 9}{\sqrt{9+36+36}} = 31.819 - 12 = 19.819 \text{ Volts}$$

**Example**: A uniform line of length 2m. with total charge 3 nc is situated coincident to the z-axis with its center point 2m. from the origin. At a point on the x-axis, 2m from the origin find $\vec{V}$ and $\vec{E}$.




Soln: l = 2m $\qquad \rho_l = \frac{Q}{l} = 1.5$ nc/m

Q = 3 nc is placed on the z-axis
E =?
V =? (2, 0, 0)

$$V = +\int \frac{\rho_l . dx}{4\pi \in_0 r}$$

$$V = +\frac{\rho_l}{4\pi \in_0} \int \frac{dr}{r}$$

r = |(2, 0, 0) – (0, 0, z)|

$$V = +\frac{\rho_l}{4\pi \in_0} \int_1^3 \frac{dx}{(4+z^2)^{1/2}}$$

$r = \sqrt{4+z^2}$

$$V = \frac{+\rho_l}{4\pi \in_0} \left[ \ln[z + \sqrt{z^2+4}] \right]_1^3$$

$$\int \frac{dx}{(x^2+a^2)^{1/2}}$$

$V = 9 x 1.5 [\ln(3+\sqrt{9+4}) - \ln(1+\sqrt{1+4})]$

Let x = a tanq

V = 9x1.5 [1.8879-1.1743]

V = 9.632 Volts

$dx = a \sec^2\theta d\theta$

$$\int \frac{a\sec^2\theta d\theta}{a(\sec^2\theta)^{1/2}}$$

$$\int \sec\theta\, d\theta = \ln(x+\sqrt{x^2+a^2})$$

$$E = \int \frac{\rho_l\, dl}{4\pi\in_0 R^2}.i_r \qquad \vec{R} = 2i_x - zi_z$$

$$E = \frac{\rho_l}{4\pi\in_0}\int \frac{\vec{R}}{|R|^3}.dl$$

$$E = \frac{\rho_l}{4\pi\in_0}\int \frac{(2i_x - zi_z)}{(4+z^2)^{3/2}}.dz$$

$$E = \frac{\rho_l}{4\pi\in_0}\left[\int_1^3 \frac{2i_x}{\left(4+z^2\right)^{3/2}}.dz - \int_1^3 \frac{zi_z dz}{\left(4+z^2\right)^{3/2}}\right]$$

Consider $\int_1^3 \frac{2dz}{(4+z^2)^{3/2}}dz$

Let z = 2 tanq

dz = 2 $\sec^2\theta d\theta$

$$= \int \frac{2\,x\,2\sec^2\theta\, d\theta}{2x\frac{3}{2}[2\sec^2\theta]^{3/2}}$$

$$= \int \frac{1}{2}.\frac{1}{\sec\theta}d\theta$$

$$= \frac{1}{2}\int \cos\theta\, d\theta$$

$$= \frac{1}{2}[\sin\theta] = 0.192$$

Consider $\int_1^3 \frac{zdz}{\left(4+z^2\right)^{3/2}}$

$4+z^2=t$

$2zdz = dt$

$$\Rightarrow \int \frac{dt}{2(t)^{3/2}} = \int \frac{1}{2}.t^{-3/2}dt$$

$$= \frac{1}{2}\,\frac{t^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} = \frac{1}{2}\,\frac{t^{-\frac{1}{2}}}{-\frac{1}{2}}$$

$$= -\frac{1}{t^{\frac{1}{2}}} = \left[-\frac{1}{\sqrt{4+z^2}}\right]_1^3$$

$$= -\left[\frac{1}{\sqrt{4+9}} - \frac{1}{\sqrt{4+1}}\right]$$

$$= -\left[\frac{1}{\sqrt{13}} - \frac{1}{\sqrt{5}}\right] = 0.17$$

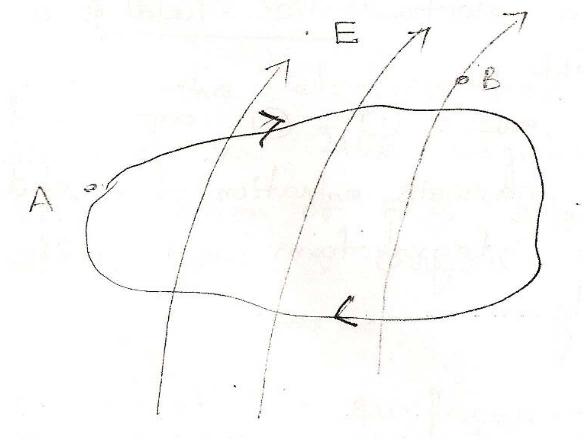
$$\therefore\ E = \frac{1.5x10^{-9}}{4\pi\, x\,\frac{10^{-9}}{36\pi}}\left[0.192i_x - 0.17i_z\right]$$

$\Rightarrow$ E = 13.5[1.192$i_x$-0.17$i_z$]

E = 2.59 $i_x$ – 2.296 $i_z$ v/m

# -: Maxwell's 2nd equation:-

$\rightarrow$The potential difference between points A and B is independent of the path taken.




$\rightarrow$ $V_{AB} = -V_{BA}$

i.e., VAB + VBA = 0

$$\int_A^B E.dl + \int_B^A E.dl = 0 \text{ (or) } \oint E.dl = 0 \rightarrow (1)$$

$\rightarrow$ This shows that the Line integaral of E along a closed path must be zero.

$\rightarrow$ Physically, this implies that no network is done in moving a charge along a closed path in an electrostatic field.

$\rightarrow$ Applying Stoke's theorem to the above eqn.

$$\oint_l E.dl = \int_S (\nabla X E).ds = 0 \text{ (or) } \nabla X E = 0 \rightarrow (2)$$

$\rightarrow$ Any vector field wahich satisfied the above equations is said to be conservative (or) irrotational field.

$\rightarrow$ Thus an electrostatic field is a conservative field.

$\rightarrow$ The equations (1) & (2) are referred to as second Maxwell's equation for static electric field in integral form and differential form respectively.

## -: Relation between E and V:-

$\rightarrow$ From the definition of potential

$$V = -\int E.dl$$

$\rightarrow$ Taking diffearentiation on both sides

dV = - E.dl

$\rightarrow$ If $E = E_x i_x + E_y i_y + E_z i_z$

& $dl = d_x i_x + d_y i_y + d_z i_z$

$\rightarrow$ Then $dv = -E_x d_x - E_y d_y - E_z d_z \rightarrow (1)$

But

$$\rightarrow\ dv = \frac{\partial v}{\partial x}.dx + \frac{\partial v}{\partial y}.dy + \frac{\partial v}{\partial z}.dz \rightarrow (2)$$

→ Comparing equations (1) & (2) for dv, we obtain

$$E_x = -\frac{\partial v}{\partial x},\ E_y = -\frac{\partial v}{\partial y} \text{ and } E_z = -\frac{\partial v}{\partial z}$$

→ Thus

$$E = -\frac{\partial v}{\partial x} i_x - \frac{\partial v}{\partial y} i_y - \frac{\partial v}{\partial z} i_z$$

$$E = -\nabla V$$

→ That is, the electric field intensity is the gradient of v. The negative sign shows that the direction of E is opposite to the direction in which V increases.

***

## -: <u>Equipotential surfaces</u>:-

→ Any surface on which the potential is the same through out is known as an Equipotential surface.

→ No work is done in moving a charge from one point to another along an equipotential surface. i.e., $V_A$-$V_B$ = 0 and hence

$$\int E.dl = 0$$

→ i.e., the Lines of force (or) flux lines are always normal to equipotential surfaces.

→ Examples of equipotential surfaces for point charge and dipole are shown in figure below.

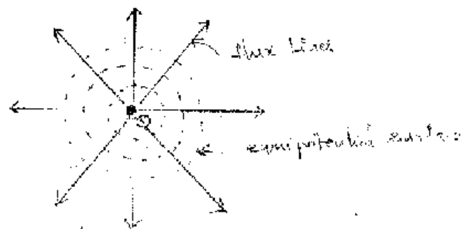



Fig: point charge.

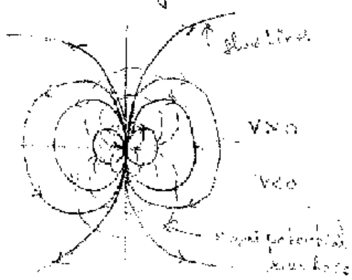



Fig: An Electric Dipole

→ The direction of E is every where normal to the equipotential surface.

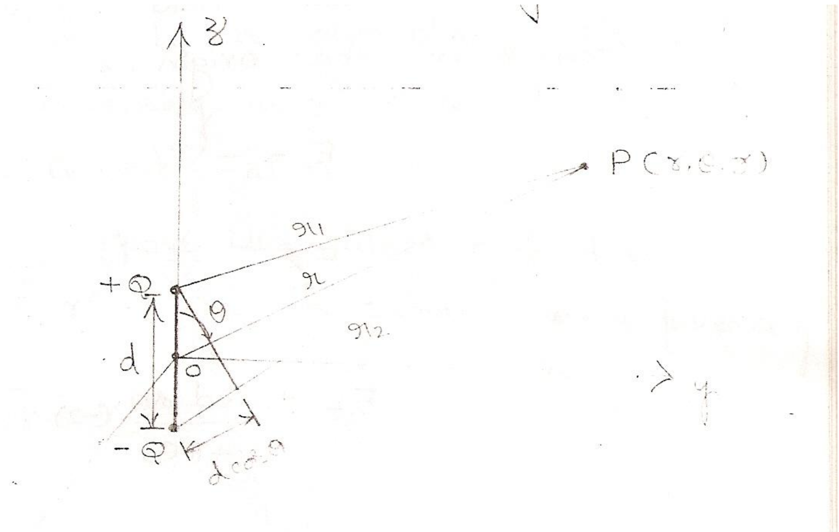
→ For static electric fields, a conductor surface is always an equipotential surface.

***

# -: Electric field intensity due to dipole:-

"An electric dipole is formed when two point charges of equal magnitude but opposite sign are separated by a small distance."

→ Consider the dipole is placed along the z-axis with its center is at the origin.




→ The potential at a point $P(r,\theta,\phi)$ is given by

$$V = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{r_1}-\frac{1}{r_2}\right]$$

$$V = \frac{Q}{4\pi\epsilon_0}\left[\frac{r_2-r_1}{r_1r_2}\right] \rightarrow (1)$$

→ Where $r_1$ and $r_2$ are the distances between P and Q and P and –Q respectively.

If r>>d,

$$r_2 - r_1 = d\cos\theta$$

$$r_1r_2 = r^2$$

→ Then equation (1) becomes as

$$V = \frac{Q}{4\pi\epsilon_0}.\frac{d\cos\theta}{r^2} \quad \frac{Q}{4\pi\epsilon_0}.\frac{d\cos\theta}{r^2} \rightarrow (2)$$

→ The electric field due to the dipole with center at the origin is

$$E = -\nabla V$$

$$E = -\left[\frac{\partial v}{\partial r}.i_r + \frac{1}{r}\frac{\partial v}{\partial \theta}.i_\theta + \frac{1}{r\sin\theta}i_\phi\right]$$

$$E = -\frac{Qd\cos\theta}{4\pi\epsilon_0}(-2)r-3.i_r + \frac{1}{r}\frac{Qd}{4\pi\epsilon_0 r^2}.\sin\theta i_\theta + 0$$

$$E = \frac{Qd\cos\theta}{2\pi \in_0 r^3} i_r + \frac{Qd\sin\theta}{4\pi \in_0 r^3} i_\theta \rightarrow (3)$$

$\rightarrow$ Since $d\cos\theta = d.i_r$ where $d = di_z$

We define P = Qd as the dipole moment then the above equations (2) & (3) becomes as

$$V = \frac{P.i_r}{4\pi \in_0 r^2} \rightarrow (4)$$

$$E = \frac{P}{4\pi \in_0 r^3}[2\cos\theta i_r + \sin\theta i_\theta] \rightarrow (5)$$

$\rightarrow$ The dipole moment P is directed from –Q to +Q.

If the dipole center is not at the origin but at $r^1$ then the potential

$$V(r) = \frac{P.(r - r^1)}{4\pi \in_0 |r - r^1|^3} \rightarrow (6)$$

$\rightarrow$ **Note:** i) Due to a point charge its electric field varies inversely as $r^2$ while it's potential field varies inversely as r.

ii) Due to a dipole the electric field varies inversely as $r^3$ while its potential varies inversely as $r^2$.

***

# -: Energy density in electrostatic fields:-

$\rightarrow$ To determine the energy present in an assembly of charges, we must first determine the amount of work necessary to assemble them.

$\rightarrow$ Suppose we wish to position three point charges $Q_1$, $Q_2$ & $Q_3$ in an initially empty space show in figurely.

$\rightarrow$ No work is required to transfer $Q_1$ from infinity to $P_1$. Because the space is initially charge free and there is no electric field.

$$w_1=0$$

$\rightarrow$ The work done in transferring $Q_2$ from infinity to $P_2$ is equal to $w_2 = Q_2V_{21}$.

$V_{21}$ is the potential at $P_2$ due to $Q_1$.

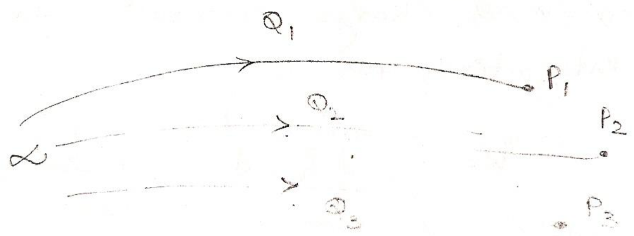
$\rightarrow$Similarly the work done in positioning $Q_3$ at $P_3$ equal to $W_3 = Q_3[V_{31}+V_{32}]$

Where $V_{31}$ and $V_{32}$ are the potentials at $P_3$ due to $Q_1$ and $Q_2$ respectively.

$\rightarrow$ Hence, the total work done in positioning the three charges is

$$W_E = W_1 + W_2 + W_3$$

$$W_E = 0 + Q_2V_{21} + Q_3[V_{31}+V_{32}] \rightarrow (1)$$




→ If the charges were positioned in reverse order,

$$W_E = W_3 + W_2 + W_1$$

$$W_E = 0 + Q_2\,[V_{23}] + Q_1\,[V_{12}+V_{13}] \rightarrow (2)$$

Where $V_{23}$ is the potential at $P_2$ due to $Q_3$. $V_{12}$ and $V_{13}$ are the potentials at $P_1$ due to $Q_2$ and $Q_3$.

→ Adding equations (1) & (2) gives

$$2W_E = Q_1(V_{12}+V_{13})+Q_2(V_{21}+V_{23})+Q_3(V_{31}+V_{32})$$

$$2W_E = Q_1V_1 + Q_2V_2 + Q_3V_3 \rightarrow (3)$$

Where $V_1$, $V_2$ and $V_3$ are the potentials at $P_1$, $P_2$ and $P_3$ respectively.

→ In general, if there are n point charges then the above equation becomes

$$WE = \frac{1}{2}\sum_{K=1}^{n} Q_K V_K \text{ Jouls} \rightarrow (4)$$

→ If, instead of point charges, the region has a continuous charge distribution, the summation becomes integration, that is

$$W_E = \frac{1}{2}\int \rho_L\, V.dl \quad \text{(Line charge)} \rightarrow (5)$$

$$W_E = \frac{1}{2}\int \rho_S\, V.ds \quad \text{(Surface charge)} \rightarrow (6)$$

$$W_E = \frac{1}{2}\int \rho_V\, V.dv \quad \text{(Volume charge)} \rightarrow (7)$$

→ Since $\rho_V = \nabla.D$ (Gauss's Law) $\frac{1}{2}\int (\nabla.D)Vdv$ (8)

Substituting in eqn (7)

$$W_E = \frac{1}{2}\int (\nabla.D)Vdv \rightarrow (9)$$

→ For any vector A and scalar V, the identity

$$\nabla.VA = A.\nabla V + V(\nabla.A)$$

$$V(\nabla.A) = \nabla.VA - A.\nabla V \rightarrow (10)$$

→ Substituting this identity in eqn (9)

$$W_E = \frac{1}{2}\int_V (\nabla.VD)dv - \frac{1}{2}\int (D.\nabla V)dv \rightarrow (11)$$

→ By applying the divergence theorem to the first term on the right-hand side of the above eqn.

$$W_E = \frac{1}{2}\oint_S (VD).ds - \frac{1}{2}\int (D.\nabla V)dv \rightarrow (12)$$

→ For point charges 'V' varies as $\frac{1}{r}$ and 'D' as $\frac{1}{r^2}$. Hence VD in the first term on the right-hand sides of above equation must vary lat least as $\frac{1}{r^3}$ while ds varies as r² consequently, the first integral must tends to zero as the surface S becomes large.

→ Hence the above eqn becomes

$$W_E = -\frac{1}{2}\int_V (D.\nabla V)dv$$

$$W_E = +\frac{1}{2}\int_V (D.E)dv \rightarrow (13)$$

Since $E = -\nabla V$

$$D = \in_0 E$$

$$W_E = \frac{1}{2}\int (D.E)dv = \frac{1}{2}\int \in_0 E^2 dv$$

→From this we can define electro static energy density in jouls/m³ as

$$\frac{dW_E}{dv} = \frac{1}{2}(D.E) = \frac{1}{2}\in_0 E^2 = \frac{D^2}{2\in_0}$$

***

# -: Convection and Conduction Currents:-

→ The current (in amperes) through a given area is the electric charge passing through the area per unit time.

i.e., $I = \frac{dQ}{dt}$

→ The current density at a given point is the current through as unit normal area at that point.

$$J = \frac{\Delta I}{\Delta S}$$

→ If the current density is not normal to the surface

$$\Delta I = J.\Delta S$$

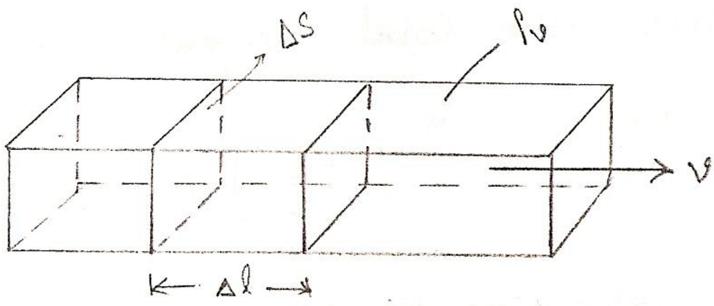
→ Thus, the total current flowing through a surface 'S' is

$$I = \int_S J.ds$$

→ Depending on how I is produced, there are different kinds of current densities: Convection current density, conduction current density, and displacement current density.

## -: Convection Current:-

→ Convection Current does not involve conductors.
→ Convection Current does not satisfy Ohm's Law.
→ It occurs when current flows through an insulating medium such as liquid, rarefied gas (or) a vaccum.
→ A beam of electrons in a vaccum tube is example for convection current.
→ Consider a filament of shown in fig. If there is a flow of charge, of density $\rho_V$ at velocity $\upsilon$, then the current from the filament is




→ $$\Delta I = \frac{\Delta Q}{\Delta t}$$

$$\Delta I = \frac{\rho_V.\Delta V}{\Delta t} = \rho_V.\Delta S.\frac{\Delta l}{\Delta t}$$

$$\Delta I = \rho_V.\Delta S.\upsilon$$

The current density J = $\frac{\Delta I}{\Delta S} = \rho_V.\upsilon$

$$J = \rho_V.\upsilon$$

The current I is the convection current and J is the convection current density in ampere/square meter.

## -: Conduction Current:-

→ Conduction current requires a conductor.
→ A conductor is characterized by a large amount of free electrons that provide conduction current due to an impressed electric field.
→ When an electric field E is applied, the force on an electron with charge –e is

$$F = -eE \rightarrow (1)$$

$\rightarrow$ If the electron with mass 'm' is moving in an electric field E with an average drift velocity $\upsilon$. Then from Newton's Law, the average change in momentum of the free electron must match the applied force (ma=-eE).

$\rightarrow$ Thus $\frac{m\upsilon}{t} = -eE$

or $\upsilon = -\frac{et}{m}E \rightarrow (2)$

Where t is the average time interval between collisions.

$\rightarrow$ If there are n electrons per unit volume, the electronic charge density is given by

$$\rho_V = -ne \rightarrow (3)$$

$\rightarrow$ Thus the conduction current density is

$$J = \rho_V \upsilon = -ne \frac{-et}{m} E$$

$$J = \frac{ne^2t}{m} E$$

Point of Ohm's law J = $\sigma E$

Where $\sigma = \frac{ne^2t}{m}$ is the conductivity of the conductor.

**<u>Note:-</u>**

$\rightarrow$ A perfect conductor cannot contain an electrostatic field with in it. Under static conditions,

$$E = 0, \rho_V = 0, V_{ab} = 0 \text{ inside a conductor.}$$

$\rightarrow$ Suppose the conductor has a uniform cross section 'S' and is of length 'l'. Then the electric field applied is uniform and its magnitude is given by

$$E = I/S$$

Since the conductor has uniform cross section

$$\therefore J = I/S$$

But J = $\sigma E$

$$J = \frac{I}{S} = \sigma E = \sigma \frac{V}{l}$$

$$\Rightarrow \frac{V}{I} = \frac{l}{\sigma S}$$

$$R = \frac{l}{\sigma S}$$

$$R = \frac{\rho_C . l}{S}$$

$\rho_C$ is the resistivity of the material

$$\rho_C = \frac{1}{\sigma}$$

# -: Linear, Isotropic and Homogeneous dielectrics:-

→ **Linear: -** A material is said to be Linear if 'D' varies linearly with 'E' and non Linear other wise.

→ **Isotropic: -** Materials for which D and E are in the same direction are said to be 'isotropic'. i.e., isotropic di-electrics are those which have the same properties in all directions.

→ **Homogeneous: -** Materials for which $\in(\text{or}\,\sigma)$ does not vary in the region being considered and is therefore the same at all points are said to be "homogeneous". They are said to be inhomogeneous when $\in$ is dependent of the space co-ordinates.

"A dielectric material is Linear if $\in$ does not change with the applied E field, homogeneous if $\in$ does not change from point to point and isotropic if $\in$ does not change with directions."

# -: Continuity Equation:-

→ Due to the principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the closed surface of the volume.

i.e., $I_{out} = \oint J.ds = -\dfrac{dQ_{in}}{dt} \rightarrow (1)$

$Q_{in}$ is the total charge enclosed by the closed surface.

→ Applying divergence theorem to the above equation

$$\oint_S J.ds = \int_V (\nabla.J)dv \rightarrow (2)$$

→ $$-\frac{dQ_{in}}{dt} = -\frac{d}{dt}\int_V \rho_V\, dv$$

$$-\frac{dQ_{in}}{dt} = -\frac{d}{dt}\int_V \frac{\partial \rho_V}{\partial t} dv \rightarrow (3)$$

→ Substituting (2) and (3) in eqn.(1)

$$\int_V (\nabla.J)dv = -\int_V \frac{\partial \rho_V}{\partial t}.dv$$

$$\nabla.J = -\frac{\partial \rho_V}{\partial t}$$

which is called the continuity of current equation.

$\rightarrow$ For steady currents, $\frac{\partial \rho_V}{\partial t} = 0$ and hence $\nabla . J = 0$ showing that the total charge leaving a volume is the same as the total charge entering it.

## -: Relaxation Time:-

$\rightarrow$ From Ohm's Law J = $\sigma E \rightarrow (1)$
$\rightarrow$ From Gauss's Law $\nabla . D = \rho_V$

$$\nabla . E = \frac{\rho_V}{\in} \rightarrow (2)$$

$\rightarrow$

$$\nabla . \sigma E = \frac{\sigma \rho_V}{\in}$$

$$\nabla . J = \frac{\sigma \rho_V}{\in} \rightarrow (3)$$

$\rightarrow$ From continuity equation $\nabla . J = -\frac{\partial \rho_V}{\in} \rightarrow (4)$

$\rightarrow$ By equating equations (3) & (4) we get

$$-\frac{\partial \rho_V}{\partial t} = \frac{\sigma \rho_V}{\in}$$

$$\frac{\partial \rho_V}{\partial t} + \frac{\sigma}{\in} \rho_V = 0 \rightarrow (5)$$

$\rightarrow$ The above equation is a Linear, homogeneous differential equation.

By separating variables

$$\frac{\partial \rho_V}{\partial t} = -\frac{\sigma}{\in} \rho_V$$

$$\frac{\partial \rho_V}{\rho_v} = -\frac{\sigma}{\in} \partial t$$

$\rightarrow$Integrating the above eqn. on both sides

$$\ln \rho_V = -\frac{\sigma}{\in} t + \ln \rho_{V_0}$$

where $\ln \rho_{V_0}$ is a constant of integration

Thus $\ln_e \frac{\rho_V}{\rho_{V_0}} = -\frac{\sigma}{\in} t$

$$\rho_V = \rho V_0 . e^{-\frac{\sigma}{\in} t}$$

$$\rho_V = \rho_{V_0} . e^{-t/tr}$$

Where

$$T_r = \frac{\in}{\sigma}$$

The time constant $T_r$ is known as the relaxation time (or) rearrangement time.

"Relaxation time is the time it takes a charge placed in the interior of a material to drop to 36.8 percent of its initial value".

# -: Poisson's and Laplace's equations:-

→Poisson's and Laplace's equations are easily lderived from Gauss's Law.

$$\nabla.D = \rho_V$$

$$\nabla.\in E = \rho_V \rightarrow (1) \qquad \because D = \in E$$

→ $$E = -\nabla V \rightarrow (2)$$

Substituting eqn (2) in (1)

$$\nabla.(-\in \nabla V) = \rho_V \rightarrow (3)$$

→ For a homogeneous medium the above eqn becomes as

$$\nabla^2 V = \frac{-\rho_V}{\in} \rightarrow (4)$$

This is known as "Poisson's equation".

→ When $\rho_V = 0$ (i.e., for a charge – free region)

The above equation becomes

$$\nabla^2 V = 0 \rightarrow (5)$$

which is known as "Laplace's equation".

→ The Laplace's equation in Cartesian, cylindrical, (or) spherical co-ordinates respectively is given by

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

→ $$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial v}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 v}{\partial \phi^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

→ $$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial v}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial v}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 v}{\partial \phi^2} = 0$$

→ Laplace's and Poisson's equations are useful in solving electrostatic field problems.

# -: Capacitance:-

→ To have a Capacitor we must have two (or more) conductors carrying equal but opposite charges.
→ This implies that all the flux Lines leaving one conductor must necessarily terminate at the surface of the other conductor.
→ The conductors are some times referred to as the plates of the capacitor.
→ The plates may be separated by free space (or) a dielectric.
→ The capacitance 'c' of the capacitor as the ratio of the magnitude of the charge on one of the plates to the potential difference between them that is

$$C = \frac{Q}{V}$$

$$C = \frac{\oint D.ds}{\int E.dl}$$

→ The negative sign before $V = -\int E.dl$ has been dropped because we are interested in the absolute value of V.
→ The capacitance C is a physical property of the capacitor and is measured in farads.

**A.Parallel – Plate Capacitor:-**

→ Consider the Parallel-plate Capacitor shown in fig. below that each of the plates has an area S and they are separated by a distance d.
→ We assume that plates l and 2, respectively, carry charges +Q and –Q uniformly distributed on them so that

$$\rho_S = \frac{Q}{S} \rightarrow (1)$$

→ An ideal Parallel-plate Capacitor is one in which the plate separation'd' is very small compared with the dimensions of the plate.
→ If the space between the plates is filled with a homogeneous di-electric with permitivity '$\in$' is given by

$$E = \frac{\rho_S}{\in}(-i_z)$$

$$E = -\frac{Q}{\in S} i_z \rightarrow (2)$$

→ Hence,

$$V = -\int_2^1 E.dl$$

$$V = -\int_0^d \left(\frac{-Q}{\in S} i_z\right).dz\, iz$$

$$V = \frac{Q}{V} = \frac{\in S}{d} \rightarrow (4)$$

$\rightarrow$ The energy stored in a capacitor is given by

$$W_E = \frac{1}{2}\int (D.E)dv$$

$$W_E = \frac{1}{2}\int \in E^2)dv$$

$$W_E = \frac{1}{2}\int \in \frac{Q^2}{\in^2 S^2} dv$$

$$W_E = \frac{\in Q^2}{2 \in^2 S^2}.s.d$$

$$W_E = \frac{Q^2}{2}\left(\frac{d}{\in S}\right) = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}VQ$$

$$W_E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

## B. <u>Coaxial Capacitor</u>:-

$\rightarrow$ Consider Length 'L' of two coaxial conductors of inner radius 'a' and outer radius 'b' (b > a) as shown in fig. below.

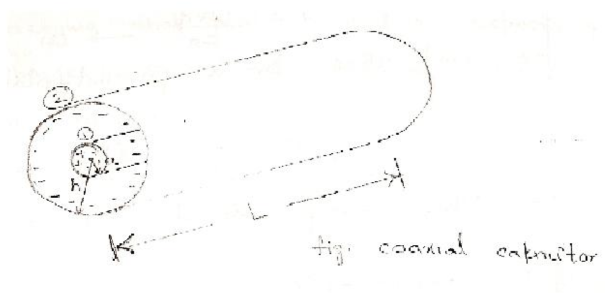


fig: coaxial capacitor

$\rightarrow$ Let the space between the conductors be filled with a homogeneous dielectric with permittivity $\in$.

$\rightarrow$We assume that conductors (1) and (2) respectively carry +Q and –Q uniformly distributed on them.

$\rightarrow$ By applying Gauss's Law to an arbitrary Gaussian cylindrical surface of radius $\rho$

(a< $\rho$ <b), we obtain

$$Q = \in \oint E.ds$$

$$Q = \in \oint_S E_\rho i_\rho . \rho d\phi dz i_\rho$$

$$Q = \in E_\rho . \rho . 2\pi . L$$

$$E_\rho = \frac{Q}{2\pi \in \rho . L}$$

Hence

$$E = \frac{Q}{2\pi \in_0 \rho_L}$$

→ Neglecting flux fringing at the cylinder ends,

$$V = -\int_1^2 E.dl$$

$$V = -\frac{Q}{2\pi \in_0 \rho_L} i_\rho . d\rho . i \rho$$

$$V = -\frac{Q}{2\pi \in_0 L} [\ln \rho]_b^a$$

$$V = \frac{Q}{2\pi \in_0 L} [\ln b - \ln a]$$

$$V = \frac{Q}{2\pi \in_0 L} \ln \left[\frac{b}{a}\right]$$

→ Thus the capacitance of a coaxial cylinder is given by

$$C = \frac{Q}{V}$$

$$C = \frac{Q}{\frac{Q}{2\pi \in_0 L} \ln \left[\frac{b}{a}\right]}$$

# C. <u>Spherical Capacitor</u>:-

→ Consider two concentric spherical conductors with the inner sphere of radius ‘a’ and outer sphere of radius b (b>a) separated by a dielectric medium with permittivity $\in$ as shown in fig. below.

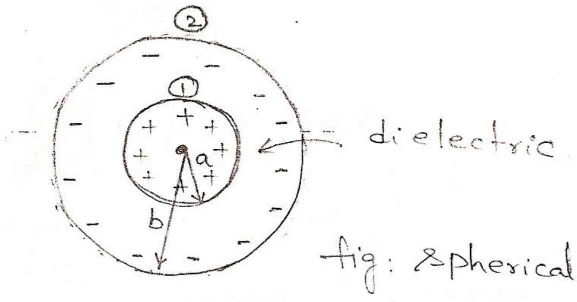



fig: spherical-capacitor

→ Assume charges +Q and –Q on the inner and outer spheres respectively.

→ By applying Gauss’s Law to an arbitrary Gaussian spherical surface of radius r (a < r < b) is given by

$$Q = \oint Q.ds$$

$$Q = \in \oint_S E.ds$$

$$Q = \in E_r.4\pi r^2$$

$$E_r = \frac{Q}{4\pi \in r^2}$$

Hence $$E = \frac{Q}{4\pi \in r^2}.i_r \rightarrow (1)$$

$\rightarrow$ The potential difference between the conductors is

$$V = -\int_2^1 E.dl = -\int_b^a \frac{Q}{4\pi r^2} ir.drir$$

$$V = \frac{Q}{4\pi \in}\left[\frac{1}{a} - \frac{1}{b}\right]$$

$\rightarrow$ Thus the capacitance of the spherical capacitor is

$$C = \frac{Q}{V}$$

$$C = \frac{Q}{\frac{Q}{4\pi \in}\left[\frac{1}{a} - \frac{1}{b}\right]}$$

$$C = \frac{4\pi \in}{\left[\frac{1}{a} - \frac{1}{b}\right]}$$

$\rightarrow$ By letting $b \rightarrow \infty, C = 4\pi \in a$, which is the capacitance of a spherical capacitor whose outer plate is infinitely large.

$\rightarrow$ The expressions for finding the resistance R and the capacitance C of an electrical system are

$$R = \frac{V}{I} = \frac{\int E.dl}{\int \sigma E.ds}$$

$$C = \frac{Q}{V} = \frac{\in \int E.ds}{\int E.dl}$$

$\rightarrow$ The product of these expressions yield's

$$RC = \frac{\in}{\sigma}$$

Which is the relaxation time $T_r$ of the medium separating the conductors.

$\rightarrow$ For a parallel-plate capacitor,

$$C = \frac{\in S}{d} \qquad R = \frac{d}{\sigma S}$$

$\rightarrow$ For a cylindrical capacitor,

$$C = \frac{2\pi \in L}{\ln b/a} \qquad R = \frac{\ln b / a}{2\pi\sigma L}$$

→ For a spherical capacitor,

$$C = \frac{4\pi \in}{\frac{1}{a} - \frac{1}{b}} \qquad R = \frac{\frac{1}{a} - \frac{1}{b}}{4\pi\sigma}$$

→ For isolated spherical capacitor,

$$C = 4\pi \in a \qquad R = \frac{1}{4\pi\sigma a}$$

# EM WAVES & TRANSMISSION LINES
# Question Bank on UNIT-I
# ELECTROSTATICS

1. State the coulomb's law in SI units and indicate the parameters used in the equations with aid of a diagram.
2. A charge $Q_1$ is at point (0, -1, 0) m. Another charge $Q_2$ is the point (0, 2, 0) m. Find the ratio $Q_2/Q_1$ resulting in zero force on a test charge at the origin. $Q_1$, $Q_2$ and the test charge are all of the same sign.
3. Point charges $Q_1$ and $Q_2$ are respectively located at (4, 0, -3) and (2, 0, 1). If $Q_2$ = 4nc, find $Q_1$ such that:
   a) The E at (5, 0, 6) has no Z-component
   b) The force on a test charge at (5, 0, 6) has no X-component.
4. Define and distinguish between the lterms Electric field, Electric Displacement and Electric flux density with necessary mathematical.
5. A point charge of 300 µC is located at point (1,2,1)m. Find the force on another point charge of 30 µC at the point (3,0,0)m and also calculate the electric field at (3,0,0)m.
6. Explain Line, Surface and Volume charge distributions.
7. A circular lring of radius 'a' carries luniform charge $\rho c/m$ and is in the xy-plane. Find the electri8c field at point (0, 0, 2) along its axis.
8. A circular disk of radius 'a' carries uniformly charged with $\rho_S$ $c/m^2$ and is z=0 plane. Find the electric field at the point (0, 0, h) along its axis.
9. State Gauss's law using divergence theorem and relate the displacement density D to the volume charge density.

10. A sphere of radius 'a' is filled with a uniform charge density $\rho_V$ C/m$^3$. Determine the electric field inside and out side the sphere.

11. In free space D = $2y^2$ix + 4xyiy – iz mc/m$^2$ find the total charges stored in the region. $1 < x < 2$, $1 < y < 2$, $-1 < z < 4$.

12. Obtain the expression for the field and potential due to an infinite line charge at any radial distance.

13. What is the potential function at point P due to point charges Q1 and Q2 at distances r1 and r2 respectively and a line charge of density $\rho_L$ c/m whose elemental charge $\rho_L.dl$ is assumed to be at distance r3 from $\rho$.

14. A uniform line of length 2m with total change 3nc is situated coincident of the Z-axis with its center point 2m fdrom the origin. At a point on the x – axis, 2m from the origin, find V and E.

15. A point change of 3nc is on the Z-axis, 2m away from the origin. Find the resultant V, E at a point on the X-axis 2m away from the origin.

16. A line change $\rho_L = 400\rho$ c/m lies along the x-axis. The surface of 3nc potential passes through the point (0, 5, 12) m, find the potential at point (2, 3, -4) m.

17. A point change of 15nc is situated at the original at the origin and another point change of -12nc is located at the point (3, 3, 3) m find E and V at the point (0,-3,-3).

18. A point change Q=222PC exists at the origin of a rectangular co-ordinate system in vacuum. Two points a and b are located at radial distance of 400 mm and 100 mm from Q along X-axis and y-axis

respectively. Find the potential difference b/w points a and b. Also the absolute potentials at a and b.

19. Obtain the expressions for the far field and potential due to a small electric dipole oriented along the Z-axis.
20. What are equipotential surfaces? Give two examples of these?
21. Establish Poisson’s and Laplace equations from Gauss’s law.
22. Explain i) Homogeneous and ii) Isotropic medium
23. Derive the expression for the electro-static energy stored in a capacitor of value C in terms of tahe total charge Q as well as the voltage V.
24. Consider two thin infinitely long concentric conducting cylinders with inner cyulinder of radius ‘a’ and outer cylinder of radius ‘b’. Derive the expression for its capacitance with air as di-electric assuming required charge distribution.

**END**